# Genomics data analysis
# 基因组学数据分析

# 分类数据(Categorical data)与富集

**李程**
北京大学生命科学学院、
生物信息中心、统计科学中心
邮件: cheng_li@pku.edu.cn

**2023年秋季助教**:
胡影绰 <hyc0322@stu.pku.edu.cn>

**课件网页**

周二下午3-5点，北大二教107 | 联系助教加入课程微信群

# 目标问题

1. 一个统计检验有哪些要素?

2. 辛普森悖论是什么，如何识别和避免?

3. 富集分析的作图有哪些种类和特点?

# 本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

# 相关性提示因果关系、机制研究假说

- 燕子低飞要落雨。
- 日落西北满天红，不是雨来就是风。
- 立秋有雨样样有，立秋无雨收半秋。
- 要知明天热不热，就看夜星密不密。

# 分类数据

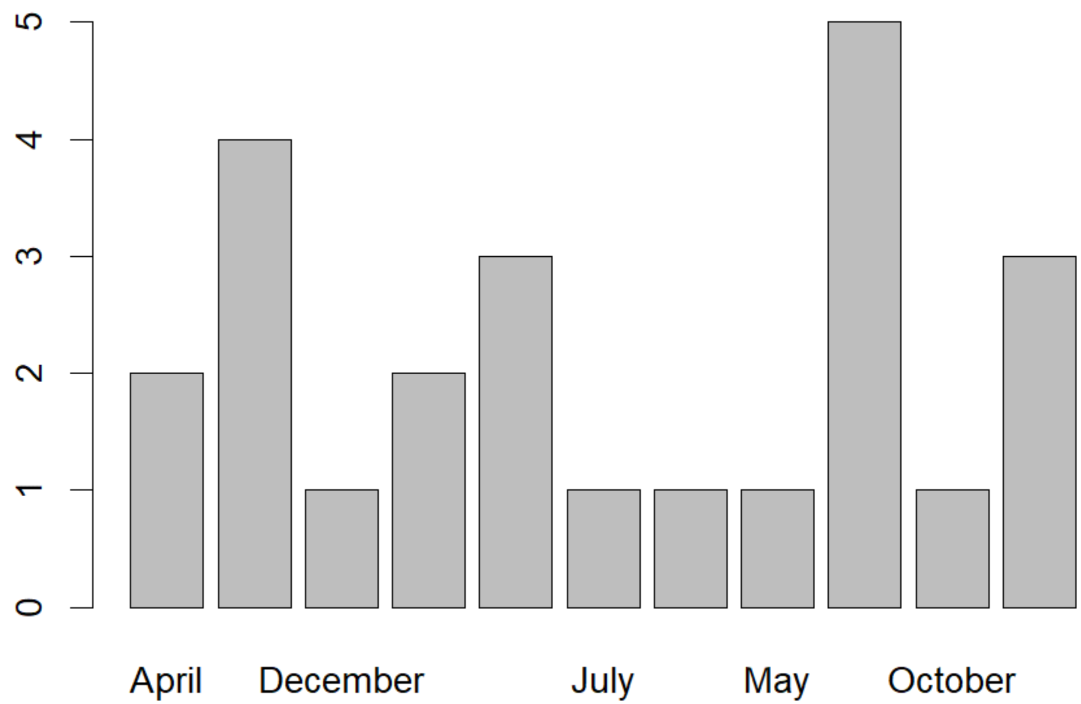
- 分类数据(categorical data)是按照现象的某种属性对其进行分类或分组而得到的反映事物类型的数据
  - 例如，按照性别将人口分为男、女两类；按照经济性质将企业分为国有、集体、私营、其他经济等。“男”、“女”，“国有”、“集体”、“私营”和“其他经济”就是分类数据。
  - 为了便于计算机处理，通常用数字代码来表述各个类别，比如，用1表示“男性”，0表示“女性”，但是1和0等只是数据的代码，它们之间没有数量上的关系和差异。
  - 对应于R语言的factor数据类型

朱喜安．统计学：湖北科学技术出版社，2014
https://baike.baidu.com/item/%E5%88%86%E7%B1%BB%E6%95%B0%E6%8D%AE

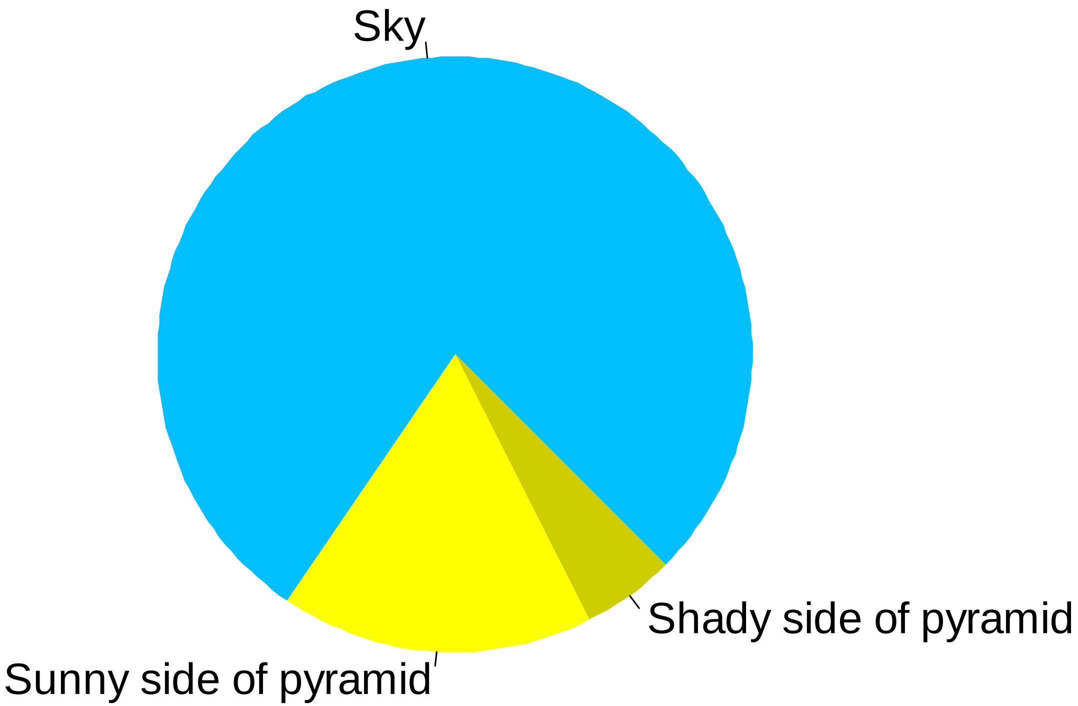
# R factor数据类型

```
mons = c("March","April","January","November","January",
 "September","October","September","November","August",
 "January","November","November","February","May","August",
 "July","December","August","August","September","November",
 "February","April")

mons

mons = factor(mons)
mons

table(mons)

plot(mons)
```

- 调用对不同数据类型的缺省plot函数
- 如何按月份或频率排序?

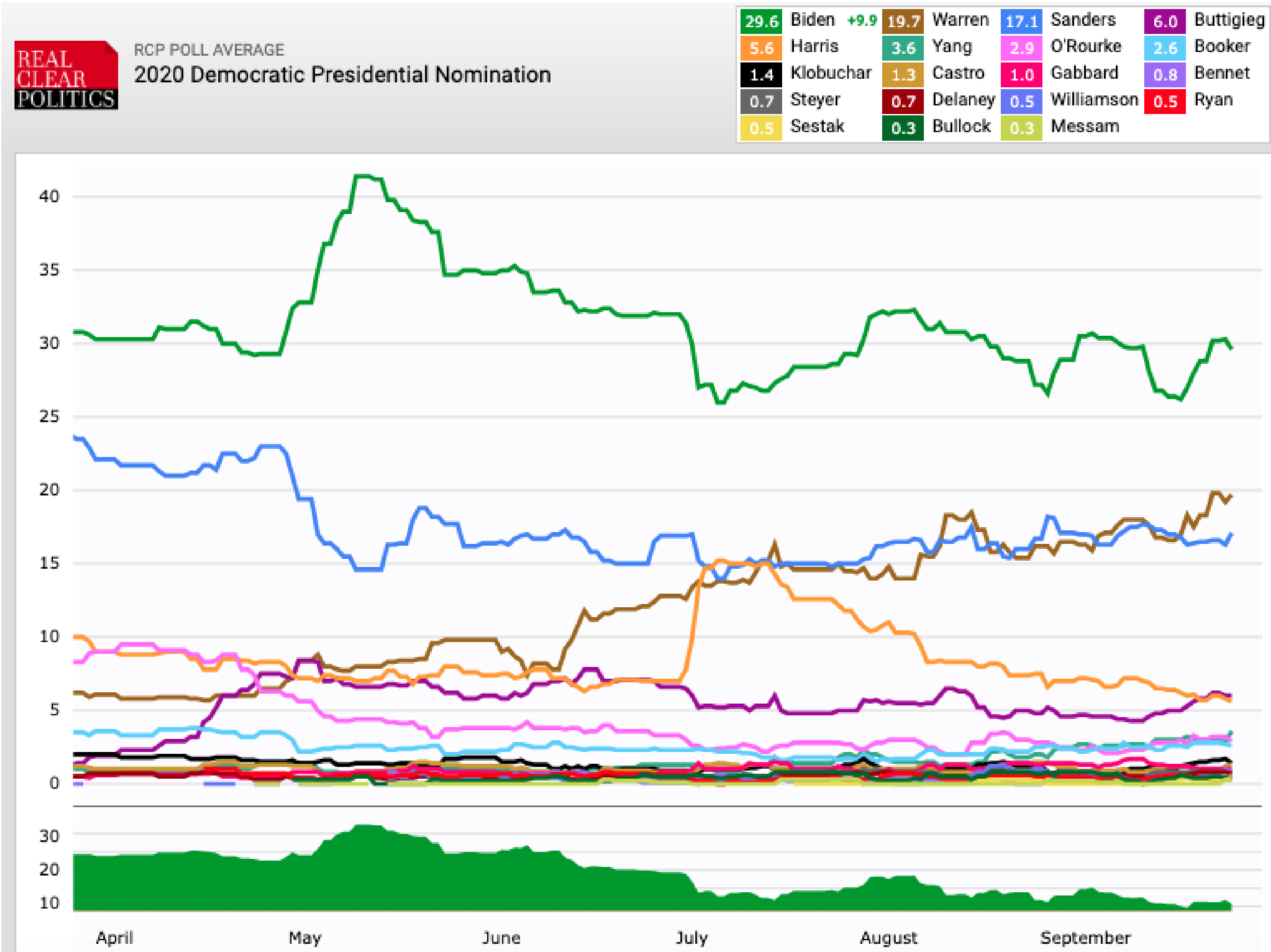


# 分类数据的常见作图：饼图



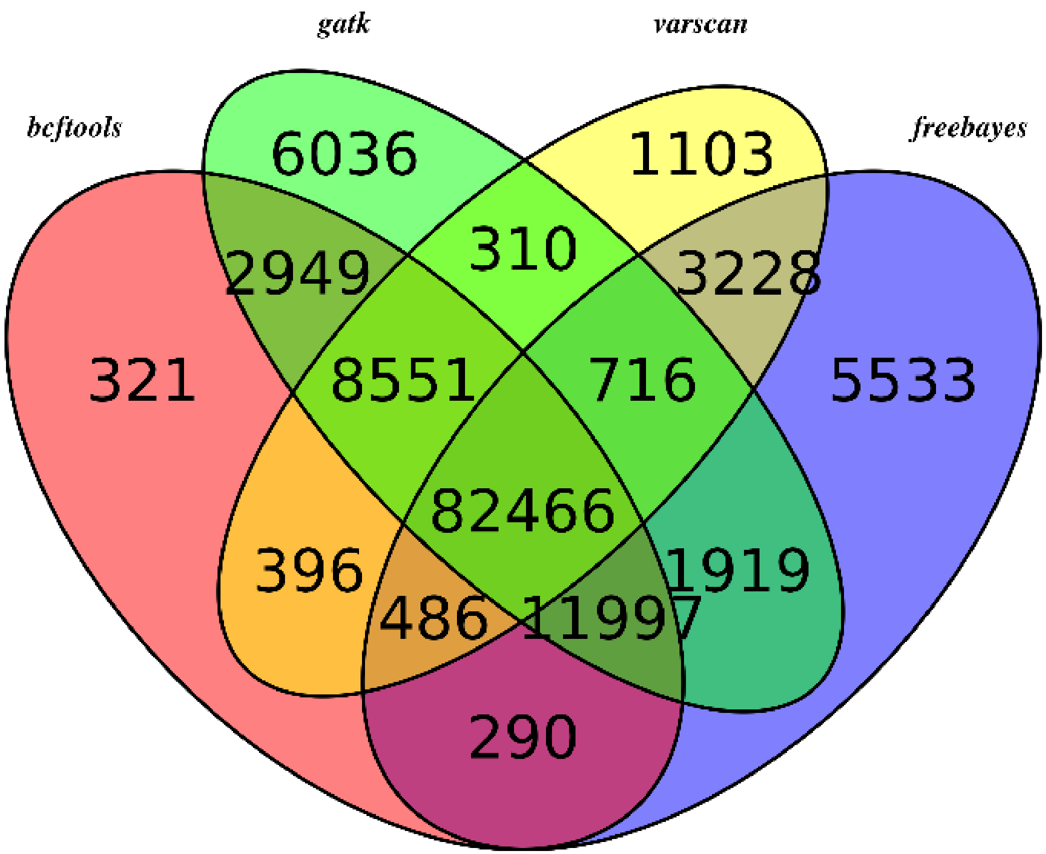

```
pie(c(Sky = 78, "Sunny side of pyramid" = 17,
       "Shady side of pyramid" = 5), init.angle = 315,
       col = c("deepskyblue", "yellow", "yellow3"),
       border = FALSE)
```

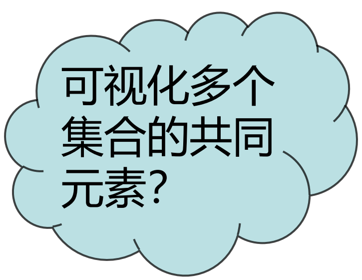
# 分类数据常转化为比例做比较




https://www.vox.com/policy-and-politics/2019/9/25/20882026/democratic-candidates-2020-polls-who-is-leading

# 韦恩图 （Venn Diagram）



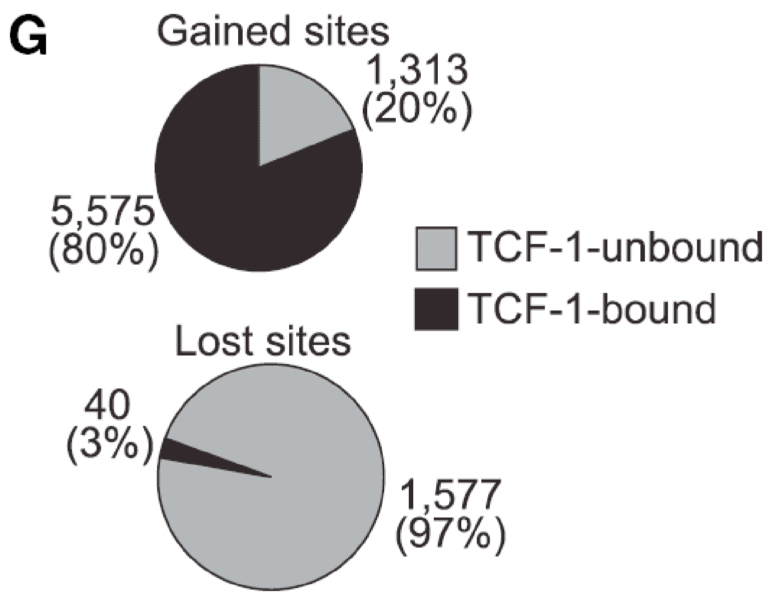




用R语言绘制韦恩图：https://www.plob.org/article/7347.html

```
install.packages("VennDiagram")
```

# 文献举例：比例的比较

**生信思路**：相关性提示因果关系：TCF-1结合的位置，染色质开放性增强较多




Cell 17 YY1 Is a Structural Regulator of Enhancer-Promoter Loops

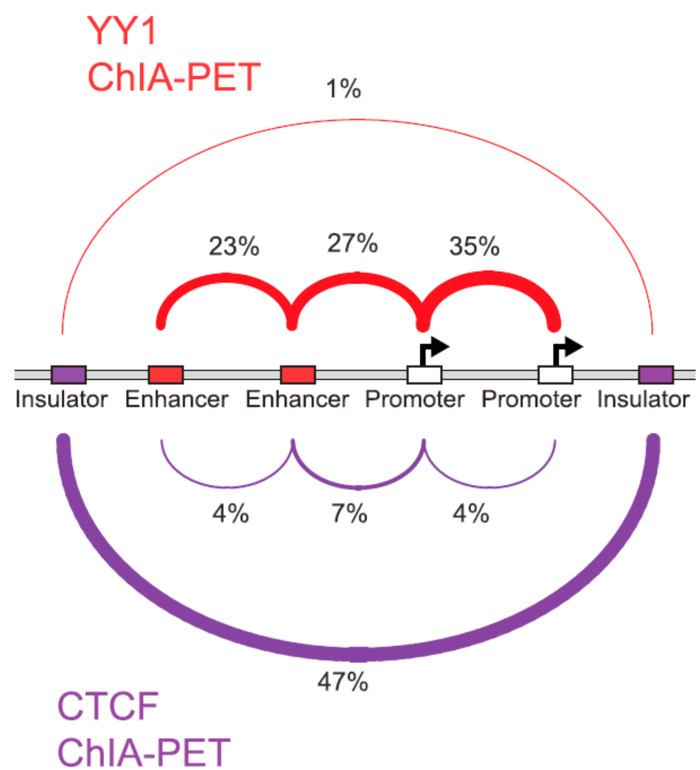



Figure 4. TCF-1 Can Bind to Nucleosomes and Create Chromatin Accessibility in Fibroblasts
(G) TCF-1 bound to 5,575 (80%) gained accessible sites in contrast to only 40 (3%) lost sites.

Immunity 18 Lineage-Determining Transcription Factor TCF-1 Initiates the Epigenetic Identity of T Cells

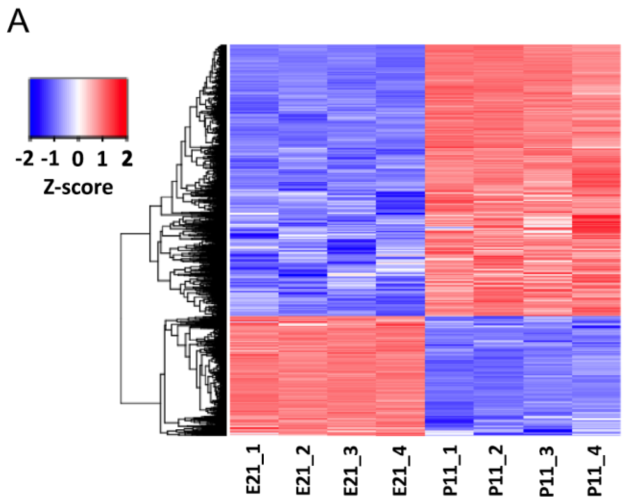
# 文献：通过染色质开放性研究视神经再生

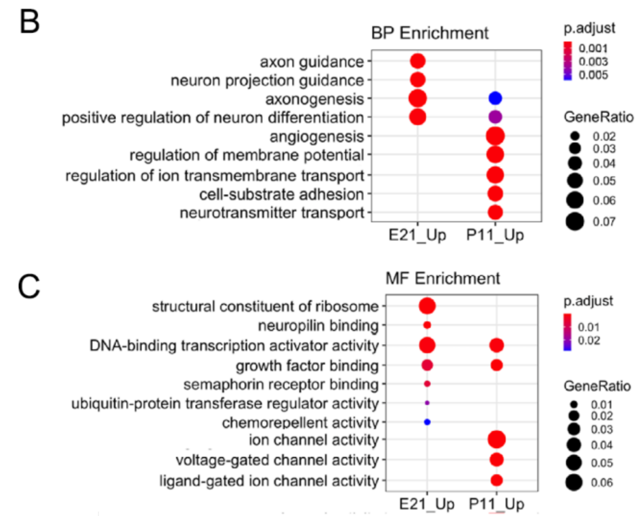
- **实验设计**：To uncover mechanisms that promote regeneration of RGC (Retinal Ganglion Cells) axons, we identified transcription factors (TF) and open chromatin regions that are enriched in rat embryonic RGCs (high axon growth capacity) compared to postnatal RGCs (low axon growth capacity).
- **组学分析：** Binding motifs for TFs such as CREB, CTCF, JUN and YY1 were enriched in the regions of the chromatin that were more accessible in embryonic RGCs.
- **功能验证：** Overexpression of CREB fused to the VP64 transactivation domain in mouse RGCs promoted axon regeneration after optic nerve injury.
- **公共数据**：The raw FASTQ files at the NCBI GEO database: GSE163564

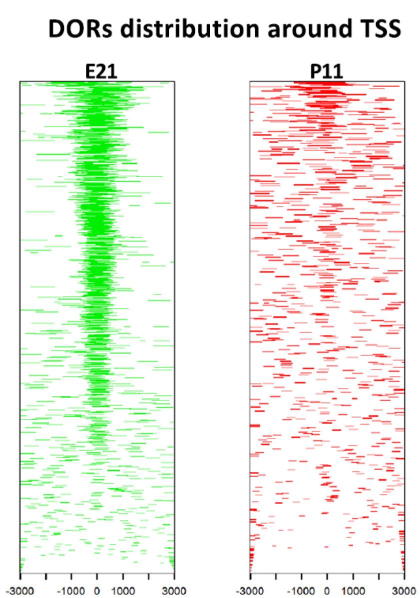
**生信思路**：比较阅读不同影响力的文献，思考实验设计、数据分析、生物发现和意义方面的异同

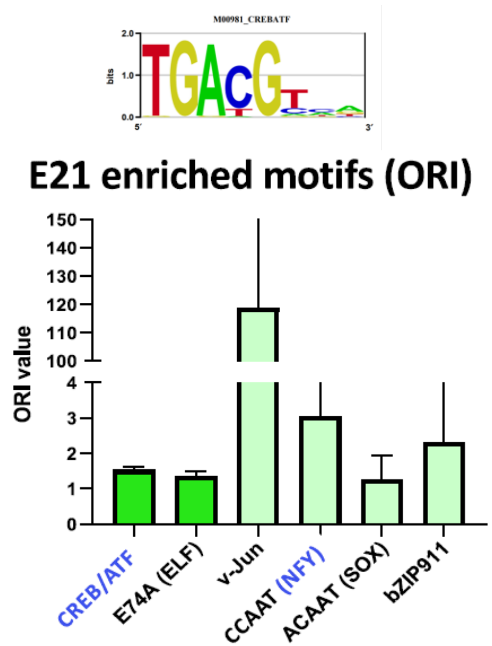
SR 21 Genome-wide chromatin accessibility analyses provide a map for enhancing optic nerve regeneration

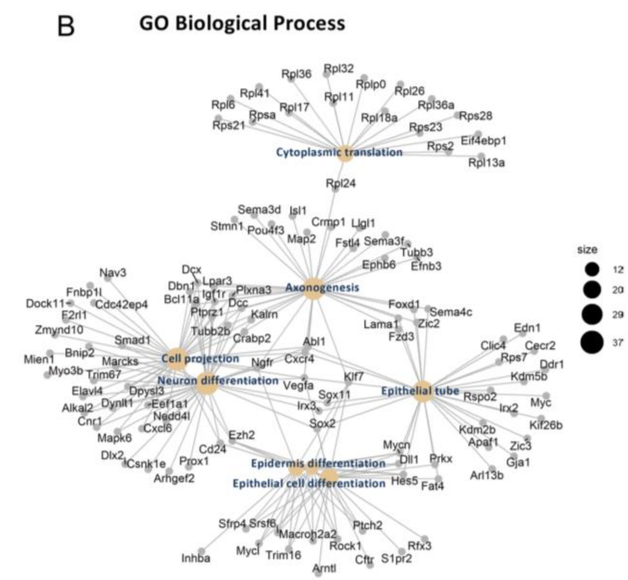
# 文献：通过染色质开放性研究视神经再生



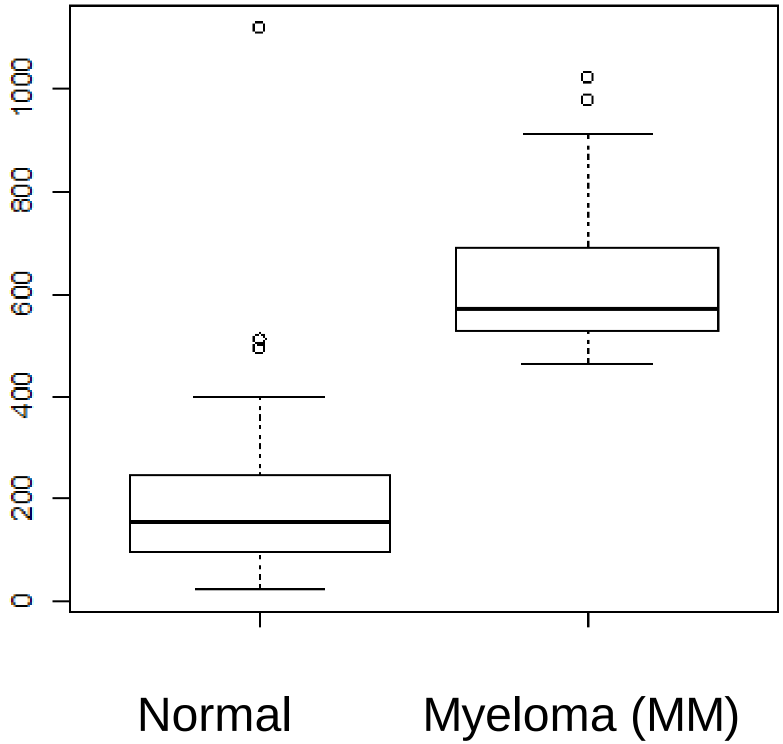



# 本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

# Differentially expressed genes

- What genes are differentially expressed between the normal and myeloma (MM) groups?




**t-test**

Test the null hypothesis that Normal and MM values come from the distributions with the same mean. (no expression difference between the two groups)

# Two-sample t-test: question

- Data points are from two groups, $x_{11}, \ldots, x_{1,n1}$ and $x_{21}, \ldots, x_{2,n2}$, which we assume are sampled independently from normal distributions:

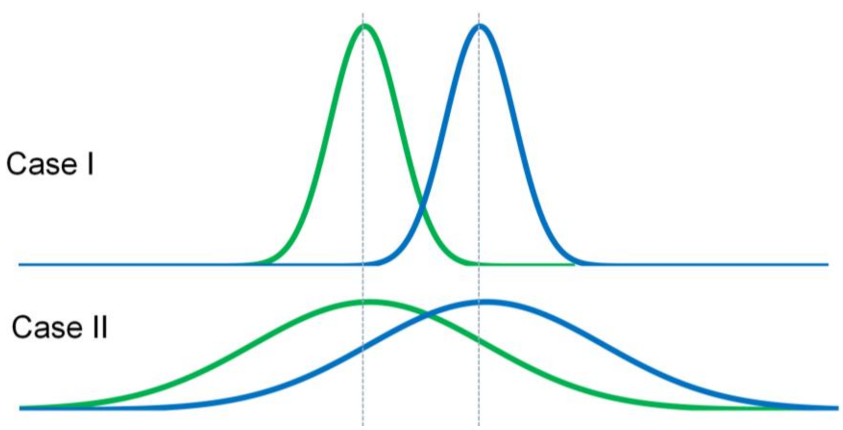
$$N(\mu_1, \sigma_1^2) \text{ and } N(\mu_2, \sigma_2^2).$$

- Question: to test the null hypothesis ($H_0$):

$$\boldsymbol{\mu_1 = \mu_2}$$

**统计检验要素:**

- 选择零假设和备选假设
- 如何设计统计量
- 统计量是什么分布




- Devise a statistic that is big when $H_1$ is true ($\mu_1 \neq \mu_2$)

# Equal variance case

- Assumes the two groups have the equal variance
  - First calculate a pooled standard deviation (SD; $S_{x1,x2}$) based on the sample SD from the two groups

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

对$X_i$ 的标准差 σ 的估计值

$$S_{X_1X_2} = \sqrt{\frac{(n_1-1)S_{X_1}^2 + (n_2-1)S_{X_2}^2}{n_1+n_2-2}}.$$

  - Use it to compute Standard Error of Difference of Mean (SEDM)

对分子的标准差的估计值

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S_{X_1X_2} \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

  - The t-statistic follows a t distribution with $n_1 + n_2 - 2$ degrees of freedom (d.f.)

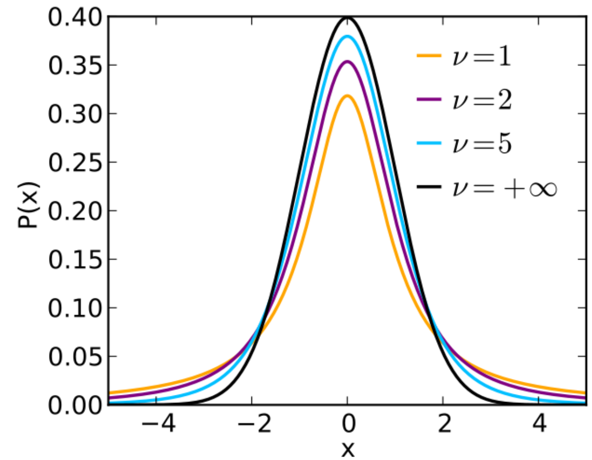
计算标准差时估计了2个均值，独立数据点减少了2个

https://en.wikipedia.org/wiki/Student%27s_t-distribution

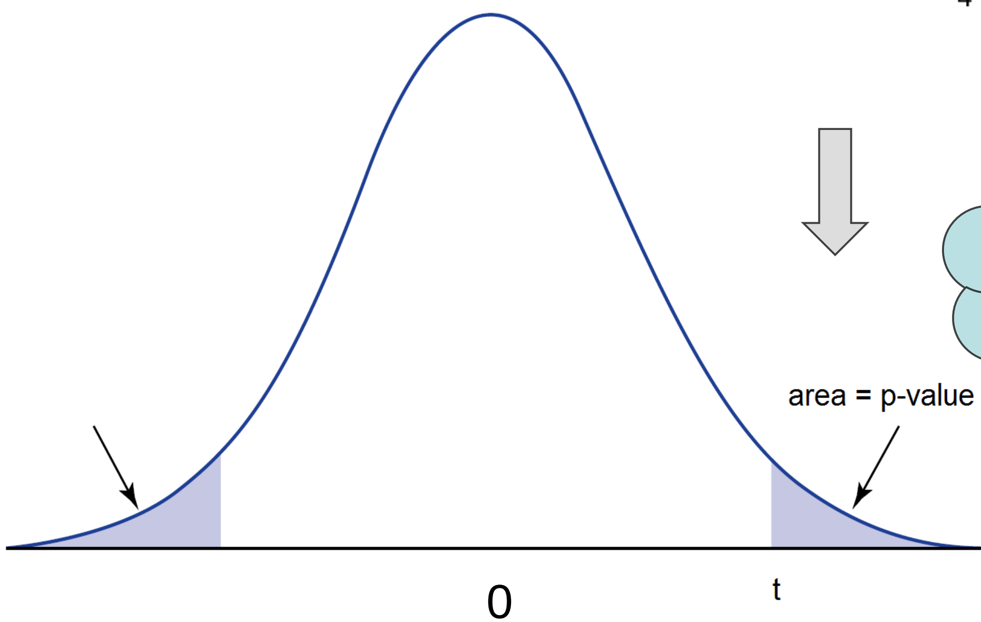
# From t-statistic to p-value

$$t = \frac{\bar{x}_2 - \bar{x}_1}{\mathrm{SEDM}}$$

设计这样的统计量 t，以方便推导出 t 的理论分布




P value：零假设下观测到极端值的概率




The distribution of the t-statistic under the null hypothesis: $\boldsymbol{\mu_1 = \mu_2}$

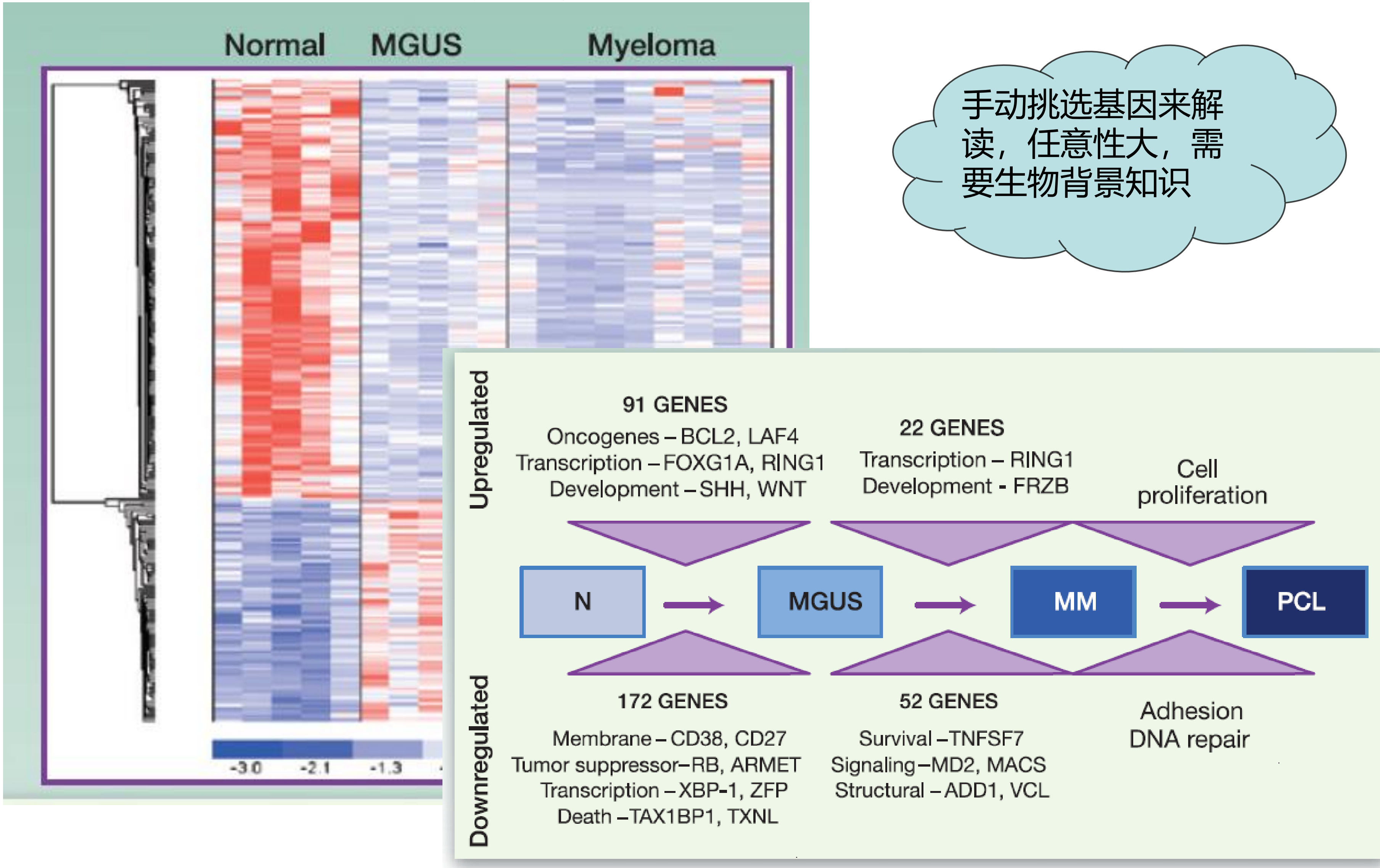
- t.test() function in R

# A myeloma expression data set

5 normal sample and 9 myeloma (MM) samples 12558 genes (rows)

| probe set | gene | Normal m4 | Normal m4 | Normal m4 | Normal m4 | Normal m43 | MM m282 | MM m331a | MM m332a | MM m333a | MM m334a | MM m353a | MM m408a | MM m423a |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 31307_at | pre-T/NK c | 28.53 | 32.61 | 29.56 | 36.55 | 33.19 | 25.1 | 32.79 | 34.3 | 35.44 | 28.48 | 29.55 | 22.28 | 28.77 |
| 31308_at | pre-T/NK c | 69.14 | 53.69 | 52.78 | 62.07 | 58.74 | 67.88 | 85.82 | 83.54 | 85.91 | 60.93 | 62.82 | 47.17 | 77.07 |
| 31309_r_a | Human bre | 16.9 | 67.7 | 27.61 | 46.16 | 51.46 | 45.62 | 35.57 | 32.62 | 35.14 | 96.18 | 45.94 | 63.2 | 38.27 |
| 31310_at | glycine rec | 67.42 | 49.55 | 55.51 | 59.57 | 68.42 | 91.06 | 91.23 | 83.66 | 76.37 | 71.23 | 74.95 | 74.04 | 100.77 |
| 31311_at | Homo sapi | 78.73 | 62.91 | 60.84 | 72.98 | 72.9 | 79.39 | 85.52 | 82.57 | 69.69 | 63.72 | 64.29 | 62.85 | 67.58 |
| 31312_at | potassium | 66.65 | 59.46 | 55.47 | 61.75 | 69.92 | 75.28 | 85.53 | 97.91 | 69.92 | 74.77 | 71.83 | 58.17 | 72.15 |
| 31313_at | mannosyl ( | 115.33 | 95.51 | 84.48 | 94.99 | 109.04 | 105.05 | 118.68 | 106.76 | 142.88 | 103.72 | 106.19 | 98.58 | 104.13 |
| 31314_at | bone morp | 71.89 | 36.24 | 41.86 | 46.99 | 45.94 | 46.67 | 67.56 | 66.14 | 53.95 | 40.97 | 47.96 | 43.63 | 53.54 |
| 31315_at | immunoglo | 103.99 | 88.27 | 83.81 | 81.81 | 254.63 | 87.12 | 99.11 | 109.56 | 86.37 | 75.03 | 74.97 | 69.02 | 97.22 |
| 31316_at | Human vac | 16.79 | 10.08 | 9.53 | 16.48 | 11.98 | 12.8 | 16.7 | 18.76 | 11.25 | 12.09 | 18.89 | 10.81 | 19.49 |
| 31317_r_a | Human unp | 316.75 | 269.61 | 254.92 | 352.61 | 342.4 | 327.12 | 366.39 | 346 | 308.43 | 279.81 | 312.4 | 318.06 | 334.27 |
| 31318_at | Stem cell f | 32.68 | 19.79 | 27.45 | 29.56 | 28.34 | 26.55 | 38.04 | 41.05 | 31.91 | 22.76 | 23.58 | 28.29 | 22.61 |
| 31319_at | Cluster Inc | 252.78 | 441.07 | 143.32 | 400.01 | 373.4 | 105.06 | 105.72 | 87.02 | 110.75 | 161.69 | 84.88 | 240.91 | 210.54 |
| 31320_at | Cluster Inc | 101.42 | 89.07 | 79.51 | 100.69 | 120.06 | 116.74 | 121.41 | 134.74 | 131.36 | 137.4 | 114.15 | 119.89 | 126.74 |
| 31321_at | Cluster Inc | 112.27 | 62.17 | 62.44 | 80.17 | 110.97 | 53.89 | 55.04 | 55.16 | 63.37 | 54.35 | 57.79 | 48.07 | 47.89 |
| 31322_at | Cluster Inc | 44.15 | 52.5 | 44.8 | 46.25 | 55.96 | 50.01 | 53.2 | 52.24 | 62.16 | 49.94 | 47.24 | 40.64 | 50.1 |
| 31323_r_a | Glutamate | 141.44 | 177.7 | 138.58 | 142.61 | 167.28 | 169.49 | 199.64 | 185.22 | 218.79 | 196.56 | 150.14 | 185.24 | 226.37 |
| 31324_at | Cluster Inc | 70.87 | 57.8 | 61.61 | 65.93 | 84.05 | 106.41 | 106.73 | 87.01 | 112.12 | 78.47 | 111 | 89.08 | 100.53 |
| 31325_at | Cluster Inc | 68.63 | 167.66 | 69.04 | 112.84 | 120.46 | 126.72 | 107.04 | 100 | 116.83 | 207.5 | 125.65 | 155.19 | 102.55 |
| 31326_at | Cluster Inc | 157.67 | 127.49 | 123.37 | 146.18 | 150.95 | 159.46 | 184.08 | 206.02 | 182.95 | 139.01 | 154.57 | 143.09 | 175.27 |
| 31327_at | Cluster Inc | 35.57 | 28.17 | 33.64 | 32.36 | 38.76 | 43.13 | 40.16 | 46.8 | 34.47 | 33.71 | 25.74 | 29.45 | 37.37 |
| 31328_at | solute carr | 61.61 | 48.23 | 50.76 | 58 | 57.58 | 57.91 | 69.91 | 72.34 | 70.29 | 54.98 | 59.74 | 45.55 | 63.02 |
| 31329_at | Human put | 12.23 | 18.91 | 15.36 | 19.99 | 21.15 | 15.9 | 20.76 | 22.26 | 16.15 | 28.86 | 13.59 | 16.06 | 23.85 |
| 31330_at | ribosomal | 108.87 | 133.3 | 89.84 | 113.02 | 147.61 | 169.87 | 156.81 | 136.47 | 153.07 | 220.54 | 220.96 | 332.11 | 183 |
| 31331_at | surfactant | 28.21 | 17.99 | 23.56 | 26.37 | 30.35 | 28.84 | 31.54 | 35.06 | 22.53 | 24.45 | 23 | 21.37 | 30.86 |
| 31332_at | RIG-like 14 | 20.77 | 18.58 | 19.03 | 18.29 | 20.86 | 23.56 | 25.11 | 24.43 | 19.3 | 28.72 | 17.27 | 23.18 | 25.35 |
| 31333_at | tolloid-like | 22.97 | 52.9 | 26.95 | 41.22 | 48.38 | 48.85 | 42.09 | 40.13 | 40.73 | 89.86 | 46.96 | 59.74 | 40.87 |
| 31334_at | G protein-c | 98.57 | 100.16 | 78.76 | 119.09 | 118.58 | 97.42 | 110.67 | 104.95 | 143.47 | 111.28 | 102.88 | 115.9 | 133.17 |
| 31335_at | clone 1900 | 65.79 | 54.3 | 54.79 | 57.23 | 60.75 | 66.98 | 72.89 | 86.97 | 76.34 | 57.65 | 59.83 | 49.21 | 70.83 |
| 31336_at | Cluster Inc | 40.97 | 26.15 | 32.55 | 26.26 | 33.73 | 36.15 | 36.03 | 34.93 | 24.12 | 26.26 | 22.55 | 23.64 | 28.11 |

# 从差异表达到生物发现




手动挑选基因来解读，任意性大，需要生物背景知识

Clinical Cancer Research 2011 Genomics in Multiple Myeloma

# A microarray data example

- Lee et al (2005) compared adipose tissue between obese and lean Pima Indians
- Samples were hybridized on HG_U95 Affymetrix microarrays
  - 12639 genes/probe sets
  - GSE2508 on the GEO database
- Selected the male samples only
  - 10 obese vs. 9 lean


GEO数据库网页： https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE2508

# A microarray data example

## Microarray profiling of isolated abdominal subcutaneous adipocytes from obese vs non-obese Pima Indians: increased expression of inflammation-related genes

Received: 10 December 2004 / Accepted: 28 April 2005 / Published online: 30 July 2005


**Abstract** *Aims/hypothesis:* Obesity increases the risk of developing major diseases such as diabetes and cardiovascular disease. Adipose tissue, particularly adipocytes, may play a major role in the development of obesity and its comorbidities. The aim of this study was to characterise, in adipocytes from obese people, the most differentially expressed genes that might be relevant to the development of obesity. *Methods:* We carried out microarray gene profiling of isolated abdominal subcutaneous adipocytes from 20 non-obese (BMI 25±3 kg/$m^2$) and 19 obese (BMI 55±8 kg/$m^2$) non-diabetic Pima Indians using Affymetrix HG-U95 GeneChip arrays. After data analyses, we measured the transcript levels of selected genes based on their biological functions and chromosomal positions using quantitative real-time PCR. *Results:* The most differentially expressed genes in adipocytes of obese individuals consisted of 433 upregulated and 244 downregulated genes. Of these, 410 genes could be classified into 20 functional Gene Ontology categories. The analyses indicated that the inflammation/immune response category was over-represented, and that most inflammation-related genes were upregulated in adipocytes of obese subjects. Quantitative real-time PCR confirmed the transcriptional upregulation of representative inflammation-related genes (*CCL2* and *CCL3*) encoding the chemokines monocyte chemoattractant protein-1 and macrophage inflammatory protein 1α. The differential expression levels of eight positional candidate genes, including inflammation-related *THY1* and *C1QTNF5*, were also confirmed. These genes are located on chromosome 11q22–q24, a region with linkage to obesity in the Pima Indians. *Conclusions/interpretation:* This study provides evidence supporting the active role of mature adipocytes in obesity-related inflammation. It also provides potential candidate genes for susceptibility to obesity.

**Electronic supplementary material** Supplementary material is available in the online version of this article at http://dx.doi.org/10.1007/s00125-005-1867-3.

皮下脂肪细胞中，炎症相关基因表达升高

Diabet 05 Microarray profiling of isolated abdominal subcutaneous adipocytes

# Differentially expressed genes

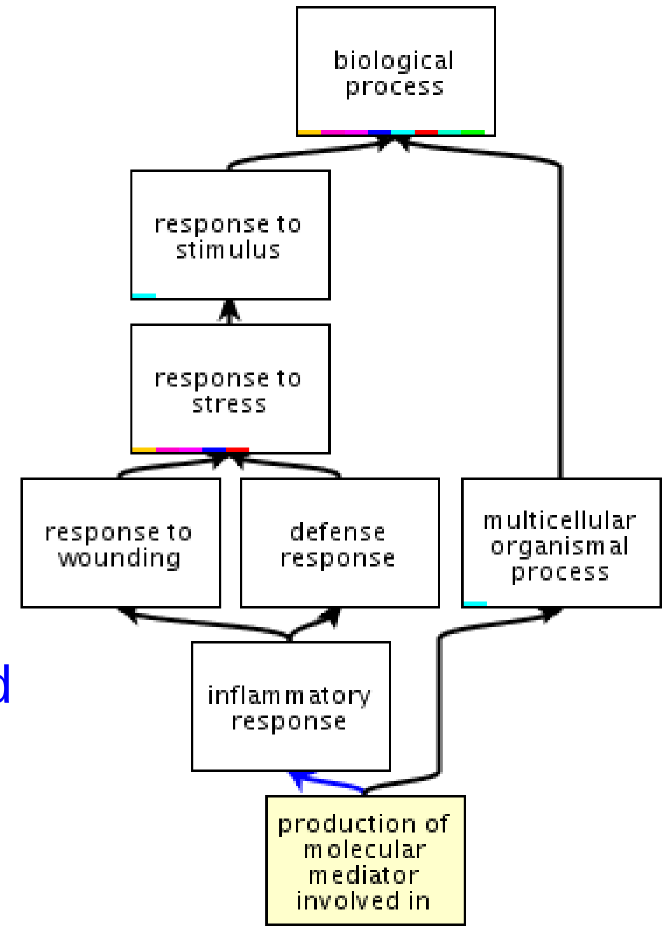
| Probe Set (Gene) | log2(obese/lean) | pvalue | adj.p |
|---|---|---|---|
| 73554_at | 1.4971 | 0.0000 | 0.0004 |
| 91279_at | 0.8667 | 0.0000 | 0.0017 |
| 74099_at | 1.0787 | 0.0000 | 0.0104 |
| 83118_at | -1.2142 | 0.0000 | 0.0139 |
| 81647_at | 1.0362 | 0.0000 | 0.0139 |
| 84412_at | 1.3124 | 0.0000 | 0.0222 |
| 90585_at | 1.9859 | 0.0000 | 0.0258 |
| 84618_at | -1.6713 | 0.0000 | 0.0258 |
| 91790_at | 1.7293 | 0.0000 | 0.0350 |
| 80755_at | 1.5238 | 0.0000 | 0.0351 |
| 85539_at | 0.9303 | 0.0000 | 0.0351 |
| 90749_at | 1.7093 | 0.0000 | 0.0351 |
| 74038_at | -1.6451 | 0.0000 | 0.0351 |
| 79299_at | 1.7156 | 0.0000 | 0.0351 |
| 72962_at | 2.1059 | 0.0000 | 0.0351 |
| 88719_at | -3.1829 | 0.0000 | 0.0351 |
| 72943_at | -2.0520 | 0.0000 | 0.0351 |
| 91797_at | 1.4676 | 0.0000 | 0.0351 |
| 78356_at | 2.1140 | 0.0001 | 0.0359 |
| 90268_at | 1.6552 | 0.0001 | 0.0421 |

多重检验
校正

Slide source: Claus-D. Mayer

# Gene Ontology (GO), 基因功能注释

- The Gene Ontology (GO) database
  - http://www.geneontology.org/
  - GO offers a relational/hierarchical database
  - Parent nodes: more general terms
  - Child nodes: more specific terms
  - At the end of the hierarchy there are genes/proteins
  - At the top there are 3 parent nodes: biological process, molecular function and cellular component




- Example
  - search the database for the GO term “inflammation”;
  - check parent and children nodes.

# GO term

## inflammatory response

### Term Information

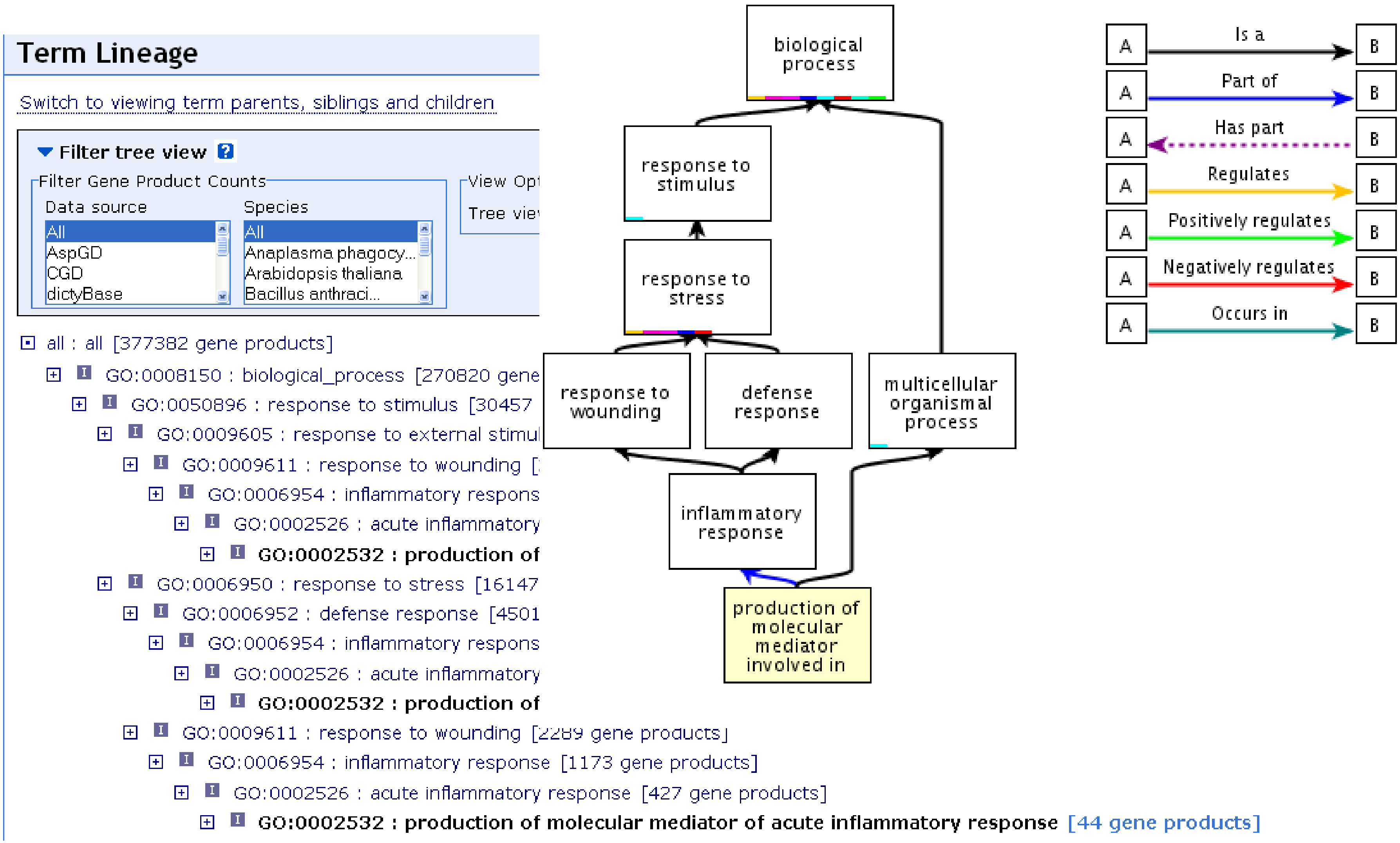
| | |
|---|---|
| **Accession** | GO:0006954 |
| **Name** | inflammatory response |
| **Ontology** | biological_process |
| **Synonyms** | Inflammation |
| **Alternate IDs** | None |
| **Definition** | The immediate defensive reaction (by vertebrate tissue) to infection or injury caused by chemical or physical agents. The process is characterized by local vasodilation, extravasation of plasma into intercellular spaces and accumulation of white blood cells and macrophages. *Source:* ISBN:0198506732, GO_REF:0000022, GOC:mtg_15nov05 |
| **Comment** | This term was improved by GO_REF:0000022. It was moved. |
| **History** | See term history for GO:0006954 at QuickGO |
| **Subset** | None |
| **Related** | Link to all **genes and gene products** annotated to inflammatory response.<br>Link to all direct and indirect **annotations** to inflammatory response.<br>Link to all direct and indirect **annotations download** (limited to first 10,000) for inflammatory response. |

Data health

http://amigo.geneontology.org/amigo/term/GO:0006954

点击下方Graph views标签 -> QuickGO

# GO term structure



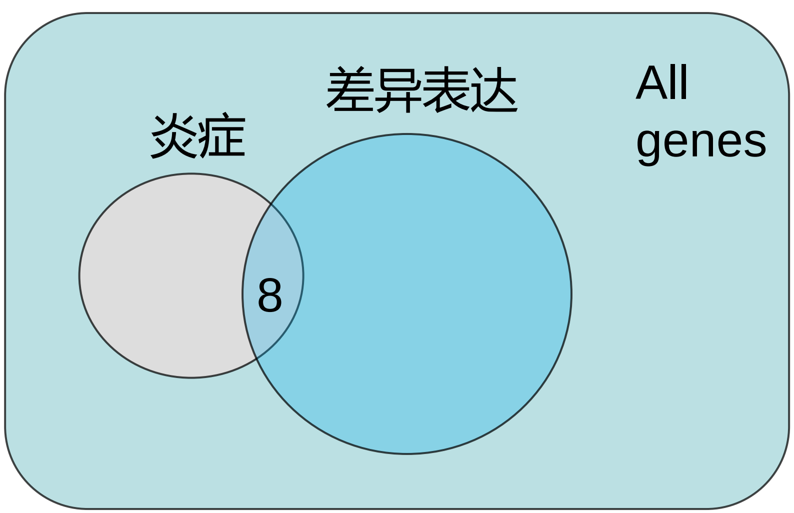
# Genes annotated by GO terms

| Probe Set ID | Gene Symbo | Gene Title | go biological process tern | go molecular function terr | log.ratio | pvalue | adj.p |
|---|---|---|---|---|---|---|---|
| 73554_at | CCDC80 | coiled-coil domain contair | --- | --- | 1.4971 | 0.0000 | 0.0004 |
| 91279_at | C1QTNF5 /// | C1q and tumor necrosis f | visual perception /// embr | --- | 0.8667 | 0.0000 | 0.0017 |
| 74099_at | --- | --- | --- | --- | 1.0787 | 0.0000 | 0.0104 |
| 83118_at | RNF125 | ring finger protein 125 | immune response /// mod | protein binding /// zinc ion | -1.2142 | 0.0000 | 0.0139 |
| 81647_at | --- | --- | --- | --- | 1.0362 | 0.0000 | 0.0139 |
| 84412_at | SYNPO2 | synaptopodin 2 | --- | actin binding /// protein bi | 1.3124 | 0.0000 | 0.0222 |
| 90585_at | C15orf59 | chromosome 15 open rea | --- | --- | 1.9859 | 0.0000 | 0.0258 |
| 84618_at | C12orf39 | chromosome 12 open rea | --- | --- | -1.6713 | 0.0000 | 0.0258 |
| 91790_at | MYEOV | myeloma overexpressed | --- | --- | 1.7293 | 0.0000 | 0.0350 |
| 80755_at | MYOF | myoferlin | muscle contraction /// blo | protein binding | 1.5238 | 0.0000 | 0.0351 |
| 85539_at | PLEKHH1 | pleckstrin homology dom | --- | binding | 0.9303 | 0.0000 | 0.0351 |
| 90749_at | SERPINB9 | serpin peptidase inhibitor, | anti-apoptosis /// signal tr | endopeptidase inhibitor a | 1.7093 | 0.0000 | 0.0351 |
| 74038_at | --- | --- | --- | --- | -1.6451 | 0.0000 | 0.0351 |
| 79299_at | --- | --- | --- | --- | 1.7156 | 0.0000 | 0.0351 |
| 72962_at | BCAT1 | branched chain aminotra | G1/S transition of mitotic | catalytic activity /// branch | 2.1059 | 0.0000 | 0.0351 |
| 88719_at | C12orf39 | chromosome 12 open rea | --- | --- | -3.1829 | 0.0000 | 0.0351 |
| 72943_at | --- | --- | --- | --- | -2.0520 | 0.0000 | 0.0351 |
| 91797_at | LRRC16A | leucine rich repeat contai | --- | --- | 1.4676 | 0.0000 | 0.0351 |
| 78356_at | TRDN | triadin | muscle contraction | receptor binding | 2.1140 | 0.0001 | 0.0359 |
| 90268_at | C5orf23 | chromosome 5 open reac | --- | --- | 1.6552 | 0.0001 | 0.0421 |

**生信思路**：GO term可以帮助理解基因功能，但如何利用它们做大规模计算分析?

# Gene function enrichment analysis

- The inflammation GO group has 96 genes (感兴趣的基因通路，如炎症)
- 1272 out of 12639 genes are differentially expressed (差异表达的基因) in this data set (10.1%)
- 问题：差异表达基因与炎症通路**是否显著相关**（8个共同基因）？
- If we randomly pick 1272 genes, how many are inflammation-related?
  - we would expect 1272 x 96 / 12639 = 9.7 genes “by chance” (we observed 8)
- A p-value can be calculated based on a 2x2 contingency table （列联表）




| | 炎症 | 非炎症 | |
|---|---|---|---|
| 差异表达 | 8 | 1264 | 1272 |
| 非差异表达 | 88 | 11 279 | 11367 |
| | 96 | 12543 | 12639 |

每个基因按两种方式分类

# R functions for count data

- Association between the counts of two categorical variables （分类变量）.
  - Regression：association between two numerical variables

- R functions to analyze tabular count data
  - prop.test() 比例比较
  - binom.test() 二项分布
  - chisq.test() 卡方检验
  - fisher.test() Fisher检验

# Tests for association with R

- Compare proportions
- Chi-square test
- Fisher's exact test

| | In GS | Not in GS |
|---|---|---|
| sig | 8 | 1264 |
| non-sig | 88 | 11 279 |

```
> x <- matrix(c(8, 1264, 88, 11279), nrow=2, byrow=T)
> x
     [,1]  [,2]
[1,]    8  1264
[2,]   88 11279
> chisq.test(x)

        Pearson's Chi-squared test with Yates' continuity correction

data:  x
X-squared = 0.15645, df = 1, p-value = 0.6924
```

**生信思路**：可以很方便地在R中计算，但要充分理解背后的统计方法，才能正确使用和解释结果

# 对多个GO term做功能富集分析

| Categories | Upregulated in obese subjects | Downregulated in obese subjects | $p$ value of category over-representation[a] | $p$ value of difference in the number of upregulated vs down regulated genes[a] |
|---|---|---|---|---|
| Inflammation/ immune response | 52 | 2 | 0.0008 | <0.00001 |
| Signalling | 25 | 17 | 0.001 | |
| Transcription regulation | 12 | 22 | 0.002 | |
| Cell cycle control/cell proliferation | 24 | 8 | | 0.002 |
| Cell adhesion | 26 | 2 | | <0.00001 |
| Transport/carrier | 18 | 14 | | |
| Structural protein/ cytoskeleton organisation | 21 | 2 | | <0.00001 |
| Apoptosis | 14 | 6 | | 0.04 |
| Protein biosynthesis/ folding | 6 | 11 | | |

Gene Ontology Biological Process classification of the 410 most differentially expressed genes as indicated by the microarray data

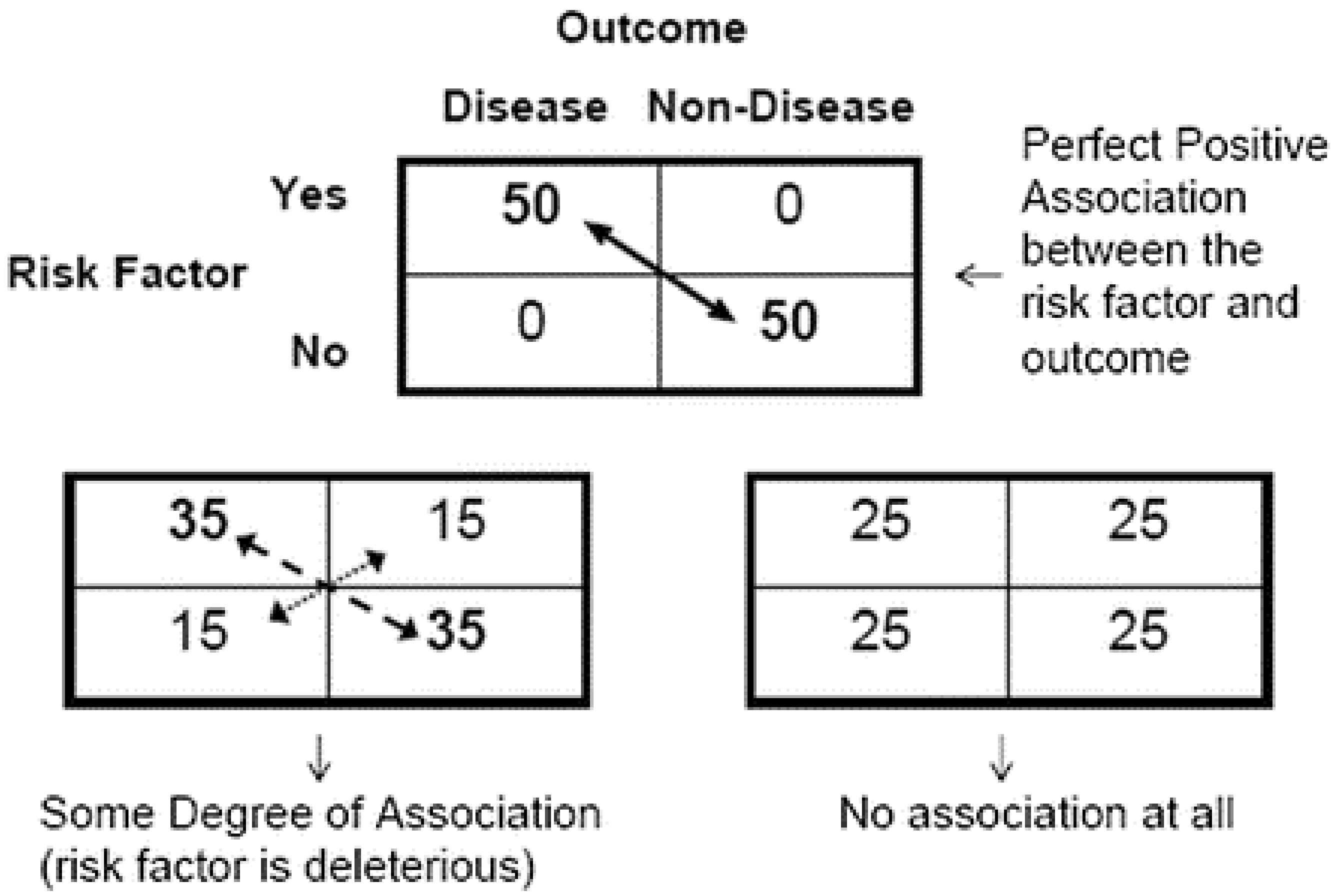
# 本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

# 统计方法类型
## (预测 / 相关性检验)

| **X变量**<br><br>**Y变量** | 数值（定量） | 类别（定性） |
| --- | --- | --- |
| 数值 | 线性回归 | t-检验，方差分析 |
| 类别 | 分类预测 | 列联表 |

# 用列联表检验分类变量的相关性




https://link.springer.com/chapter/10.1007/978-3-319-02402-8_6

# 2x2 Contingency Table （列联表）

- The table shows the data from a study of 91 patients who had a myocardial infarction (Snow 1965).
- One variable is **treatment** (propranolol versus a placebo), and the other is **outcome** (survival for at least 28 days versus death within 28 days).

**临床问题：**新药是否比安慰剂显著好？（结果是否由随机因素造成）

| Treatment | OUTCOME: Survival for at least 28 days | OUTCOME: Death | Total |
|---|---|---|---|
| Propranolol（心得安） | 38 | 7 | 45 |
| Placebo | 29 | 17 | 46 |
| Total | 67 | 24 | 91 |

Slide source: Mingqian Duan

# The hypotheses for two-way tables

- The null hypothesis $H_0$ is simply that ***there is <u>no</u> association*** between the row and column variables.
  - The method of treating the myocardial infarction patients did not influence the proportion of patients who survived for at least 28 days.
- The alternative hypothesis $H_a$ is that ***there <u>is</u> an association*** between the two variables.
  - It doesn’t specify a particular causal direction
  - Assume the outcome (survival time) depends on the treatment.

**统计检验思路**：

1. 把无效果或不感兴趣的结论作为零假设
2. 设计统计量，计算零假设下产生观测数据或更极端数据的概率（p-value）
3. 如果这个概率很小，则推翻零假设

（注意：有可能零假设、统计模型或数据点错了）

# Calculation of Expected Cell Count

- The expected cell counts under $H_0$

$$E_{1,1} = \frac{Row_1\ Total \times Column_1\ Total}{Study\ Total}$$

按总体的列比例，
把每一行总数分配
到各列

- 考虑所有病人，长期生存和短期生存人数的比例约是3：1
- 如果零假设成立（用药不影响预后生存期），每行中的计数都应该是这个比例

| | | OUTCOME | | |
|---|---|---|---|---|
| | | Survival for at least 28 days | Death | Total |
| Treatment | Propranolol | 33.13 | 11.87 | 45 |
| | Placebo | 33.87 | 12.13 | 46 |
| | Total | 67 | 24 | 91 |

# Calculation of Expected Cell Count

**Observed cell counts：**

| | | OUTCOME | | |
|---|---|---|---|---|
| | | Survival for at least 28 days | Death | Total |
| Treatment | Propranolol | 38 | 7 | 45 |
| | Placebo | 29 | 17 | 46 |
| | Total | 67 | 24 | 91 |

**Expected cell counts**（零假设下的预期结果）：

| | | OUTCOME | | |
|---|---|---|---|---|
| | | Survival for at least 28 days | Death | Total |
| Treatment | Propranolol | 33.13 | 11.87 | 45 |
| | Placebo | 33.87 | 12.13 | 46 |
| | Total | 67 | 24 | 91 |

# 统计量：Chi-Square ( $\chi^2$) Test Statistic

Calculation of the chi-square ($\chi^2$) value

$$\chi^2 = \sum\left[\frac{(O-E)^2}{E}\right]$$

$$= \frac{(38-33.13)^2}{33.13} + \frac{(7-11.87)^2}{11.87} + \frac{(29-33.87)^2}{33.87} + \frac{(17-12.13)^2}{12.13}$$

$$= 5.38$$

The process of comparing the observed counts with the expected counts is called a **goodness-of-fit test （拟合优度检验）**

$$df = (R-1)(C-1) = 1$$

$$p\text{-}value < 0.025$$

**结果解读：**如果零假设成立，$\chi^2$统计量近似服从$\chi^2$分布，在大概率情况下，统计量将取值小，p-value不显著

Interpretation:

The results noted in this $2 \times 2$ table are statistically significant.

That is, it is highly improbable (only 1 chance in about 50 of being wrong) that the difference is observed under $H_0$. Therefore the investigator can reject the null hypothesis of independence and accept the alternative hypothesis that propranolol does affect the outcome of myocardial infarction.

# The Chi-Square ( $\chi^2$) Test p-value

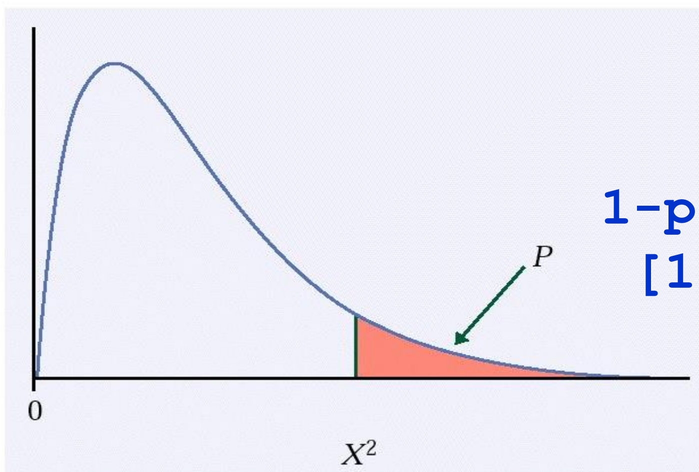
## CHI-SQUARE TEST FOR TWO-WAY TABLES

The null hypothesis $H_0$ is that there is no association between the row and column variables in a two-way table. The alternative is that these variables are related.

If $H_0$ is true, the chi-square statistic $X^2$ has approximately a $\chi^2$ distribution with $(r-1)(c-1)$ degrees of freedom.

The $P$-value for the chi-square test is

$$P(\chi^2 \geq X^2)$$

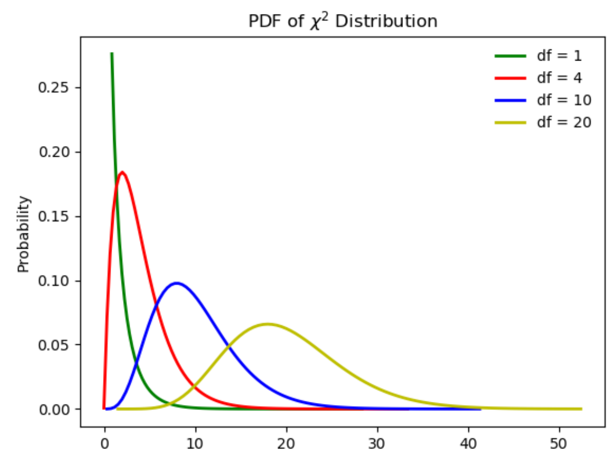



where $\chi^2$ is a random variable having the $\chi^2$(df) distribution with df $= (r-1)(c-1)$.

**Definition, pg 597**
***Introduction to the Practice of Statistics, Fifth Edition***


如果$\chi^2$统计量大，则p-value显著，说明零假设下产生这样极端的数据的概率很小，所以推翻零假设

```
1-pchisq(5.38,1)
 [1] 0.02036888
```



# 使用Chi-Square ( $\chi^2$) 检验的注意事项

The chi-square statistic is a measure of how much the observed cell counts in a two-way table differ from the expected cell counts under $H_0$.

$\chi^2$ **= Σ(observed count – expected count)$^2$ / expected count**

**Degrees of freedom (*df*) = (*r* –1) x (*c* – 1)**

- *r* is the number of rows, *c* is the number of columns, and the sum (Σ) is over all the *r* x *c* cells in the table.
- For 2 x 2 tables, need all the expected cell counts $\geq$ 5.
- For tables larger than 2 x 2, need the average of the expected cell counts is $\geq$ 5 and the smallest expected cell count is $\geq$ 1

# What do we do if the expected values in any of the cells in a 2x2 table is below 5?

| | men | women | total |
|---|---|---|---|
| dieting | 1 | 9 | 10 |
| not dieting | 11 | 3 | 14 |
| totals | 12 | 12 | 24 |

每个格子数值小时，$\chi^2$统计量不近似服从$\chi^2$分布，会导致错误P值

We guess that the proportion of dieting individuals is higher among women than among men, and we want to test whether any difference of proportions that we observe is significant.

课后阅读内容：Fisher精确检验

# Fisher's exact test（Fisher精确检验）

| | men | women | total |
|---|---|---|---|
| dieting | 1 | 9 | **10** |
| not dieting | 11 | 3 | 14 |
| totals | 12 | 12 | **24** |

The question we ask about these data is: knowing that 10 of these 24 teenagers are dieters, what is the probability that these 10 dieters would be so unevenly distributed between the girls and the boys?

• If we were to choose 10 of the teenagers at random (no relation to dieting), what is the probability that 9 of them would be among the 12 girls, and only 1 from among the 12 boys?

--Fisher's exact test uses hypergeometric distribution （超几何分布） to calculate the "exact" probability of obtaining such set of the values.

# Fisher's exact test

| | men | women | total |
|---|---|---|---|
| dieting | *a* | *c* | ***a + c*** |
| not dieting | *b* | *d* | *b + d* |
| totals | *a + b* | *c + d* | ***n*** |

R.A. Fisher showed that the probability of obtaining any such set of values was given by the hypergeometric distribution:

$$p = \binom{a+b}{a}\binom{c+d}{c} \Big/ \binom{n}{a+c}$$

$$= \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{n!a!b!c!d!}$$

# Our example

As extreme as observed

More extreme than observed

| | men | women | total |
|---|---|---|---|
| dieting | 1 | 9 | 10 |
| not dieting | 11 | 3 | 14 |
| totals | 12 | 12 | 24 |

| | men | women | total |
|---|---|---|---|
| dieting | 0 | 10 | 10 |
| not dieting | 12 | 2 | 14 |
| totals | 12 | 12 | 24 |

$$p = \frac{10!14!12!12!}{24!1!9!11!3!} = 0.00134$$

```
> dhyper(1,12,12,10)
[1] 0.001346076
```

$$p = \frac{10!14!12!12!}{24!0!10!12!2!} = 0.00003$$

```
> dhyper(0,12,12,10)
[1] 3.36519e-05
```

**dhyper(x, m, n, k)**
x：取到的白球
m：白球总数（男）
n：黑球总数（女）
k：取球总数（节食）

p-value is the probability of observing data as extreme or more extreme if the <u>null hypothesis</u> is true. So the p-value is this problem is 0.00137, rejecting $H_0$ (no association between gender and dieting).

# R fisher.test function

```
> x <- matrix(c(1, 9, 11, 3), nrow=2, byrow=T)
> x
     [,1] [,2]
[1,]    1    9
[2,]   11    3
> fisher.test(x)

        Fisher's Exact Test for Count Data

data:  x
p-value = 0.002759
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.0006438284 0.4258840381
sample estimates:
odds ratio
0.03723312
```

**双侧检验：**

```
>dhyper(1,12,12,10)+dhyper(0,12,12,10)
+dhyper(10,12,12,10)+dhyper(9,12,12,10)
[1] 0.002759456
```

```
> fisher.test(x, alternative="less")

        Fisher's Exact Test for Count Data

data:  x
p-value = 0.00138
alternative hypothesis: true odds ratio is less than 1
95 percent confidence interval:
 0.0000000 0.3260026
sample estimates:
odds ratio
0.03723312
```

**单侧检验：**

```
> dhyper(1,12,12,10)+
dhyper(0,12,12,10)
[1] 0.001379728
```

# More about Fisher’s Exact Test

- Used when one or more of the expected counts in a contingency table is small (<2).
  - There's really no lower bound on the amount of data that is needed for Fisher's Exact Test. You can use Fisher's Exact Test when one of the cells in your table has a zero in it.
  - Fisher's Exact Test is also very useful for highly imbalanced tables. If one or two of the cells in a two by two table have numbers in the thousands and one or two of the other cells has numbers less than 5, you can still use Fisher's Exact Test.

- Fisher's Exact Test is based on exact probabilities from a specific distribution (the hypergeometric distribution).
  - Fisher's Exact Test has no formal test statistic and no critical value, and it only gives you a p-value.

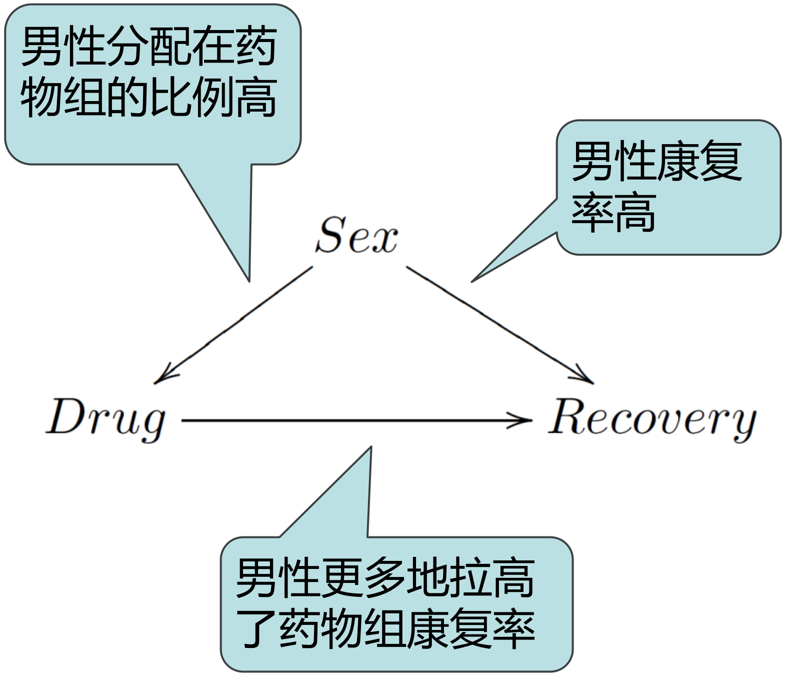
- The Chi-square test relies on a large sample approximation. Therefore, you may prefer to use Fishers Exact test in situations where a large sample approximation is inappropriate. （保险做法：两种检验都做一下，比较结果）

# 本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

# **Stat Pitfalls:** Yule-Simpson悖论

| 合并表 | 康复 | 未康复 | 康复率 |
|---|---|---|---|
| 吃药 | 20 | 20 | 50% |
| 安慰剂 | 16 | 24 | 40% |
| 男性 | 康复 | 未康复 | 康复率 |
| 吃药 | 18 | 12 | 60% |
| 安慰剂 | 7 | 3 | 70% |
| 女性 | 康复 | 未康复 | 康复率 |
| 吃药 | 2 | 8 | 20% |
| 安慰剂 | 9 | 21 | 30% |




**实验设计和分析思路**：是否充分考虑到第三方混杂因素(confounding factor，如性别)的影响？如何检查？

图片来源：丁鹏《因果推断简介》

# 研究生录取是否有性别歧视？

UC Berkeley 的统计学家**Peter Bickel** 教授1975 年在Science 上发表文章，报告了Berkley 研究生院男女录取率的差异：

- 总体上，男性的录取率高于女性
- 然而按照专业分层后，女性的录取率却高于男性
- 女性偏好申请的专业申请人数多、录取率低

专业

性别 - - - - - -> 录取率

录取率

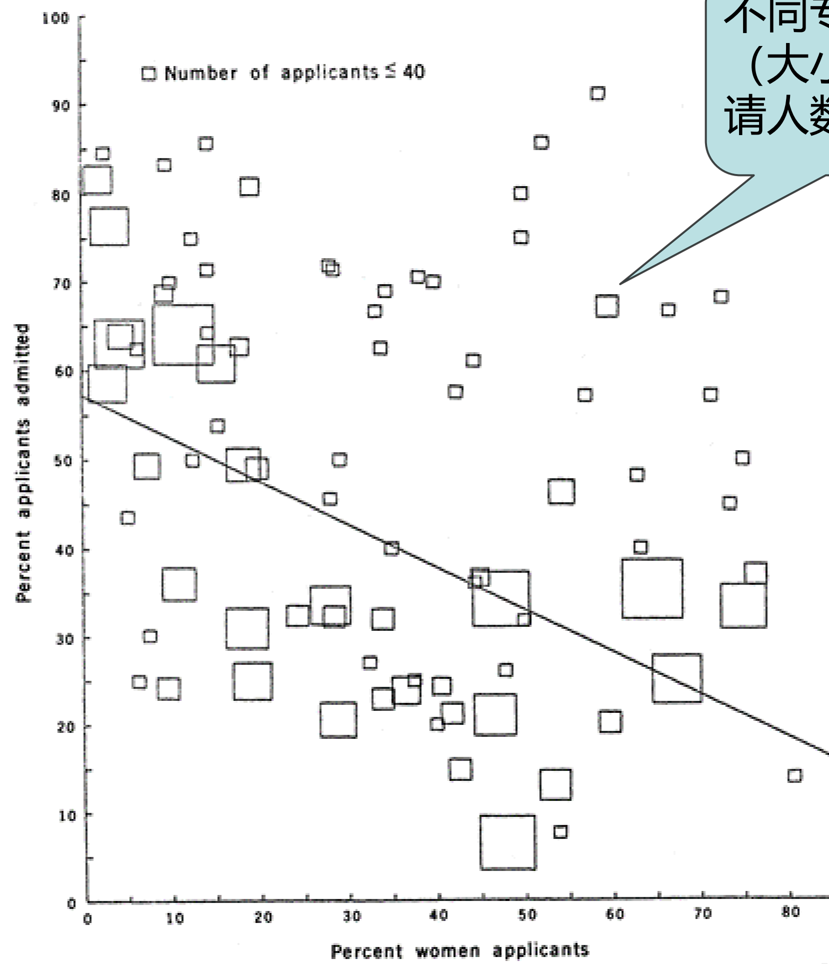



Fig. 1. Proportion of applicants that are women plotted against proportion of applicants admitted, in 85 departments. Size of box indicates relative number of applicants to the department.

不同专业（大小为申请人数）

申请者中女性比例

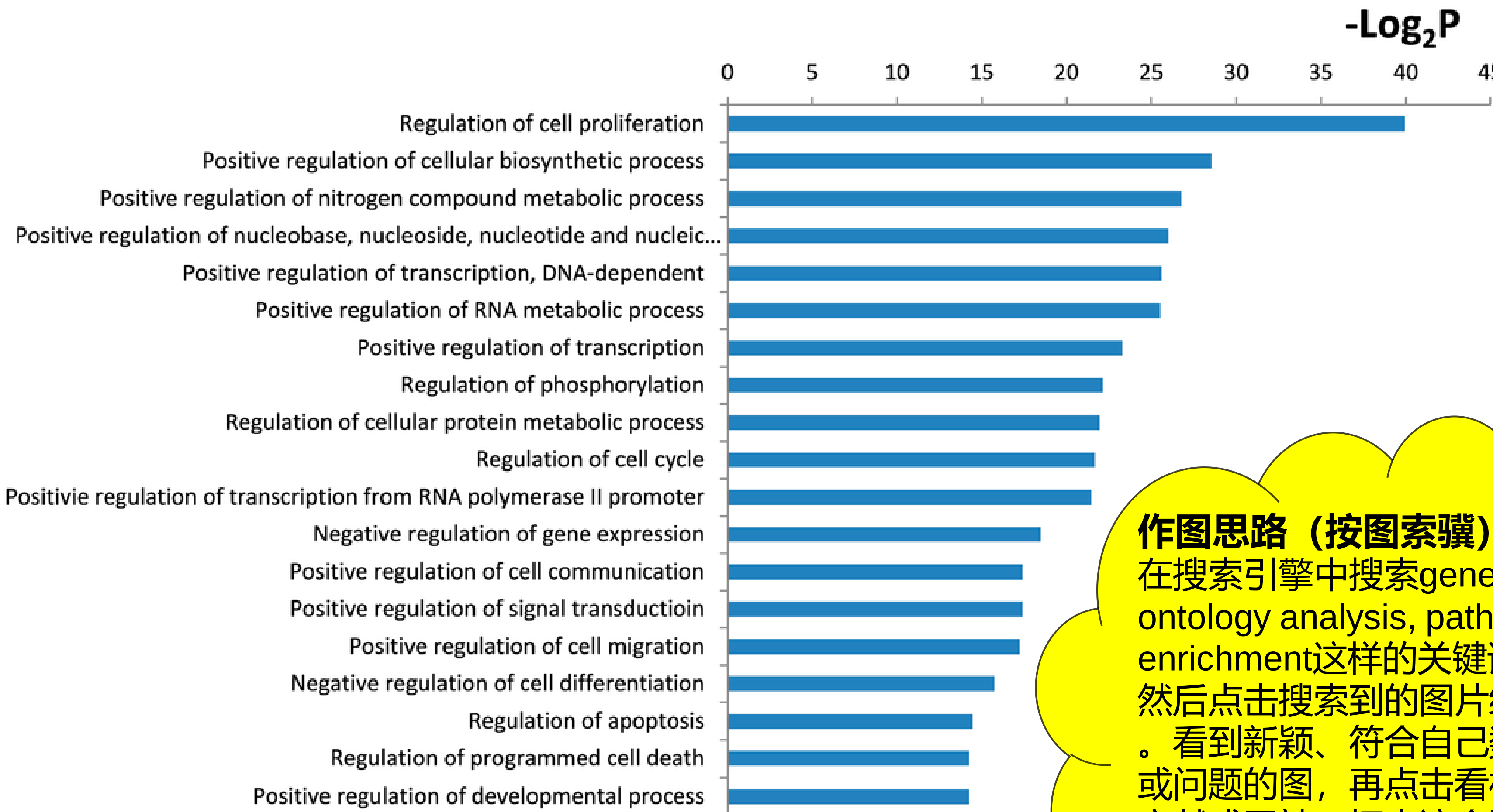
Bickel,P.J. et al, 1975, Sex Bias in Graduate Admissions: Data from Berkeley,

主页：[https://statistics.berkeley.edu/people/peter-bickel](https://statistics.berkeley.edu/people/peter-bickel)

# 本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

# **Stat Graphs**: Gene function enrichment




**作图思路（按图索骥）：**
在搜索引擎中搜索gene ontology analysis, pathway enrichment这样的关键词，然后点击搜索到的图片结果。看到新颖、符合自己数据或问题的图，再点击看相关文献或网站，探索这个图是怎么做出来的。

基因集合的功能富集、通路和网络图
https://mp.weixin.qq.com/s/ZFqfqwmXa5gfl8j7BY49Zg

# P-value plot

**优点:**

- 简洁；图比数据表能更直观、快速地传达信息。

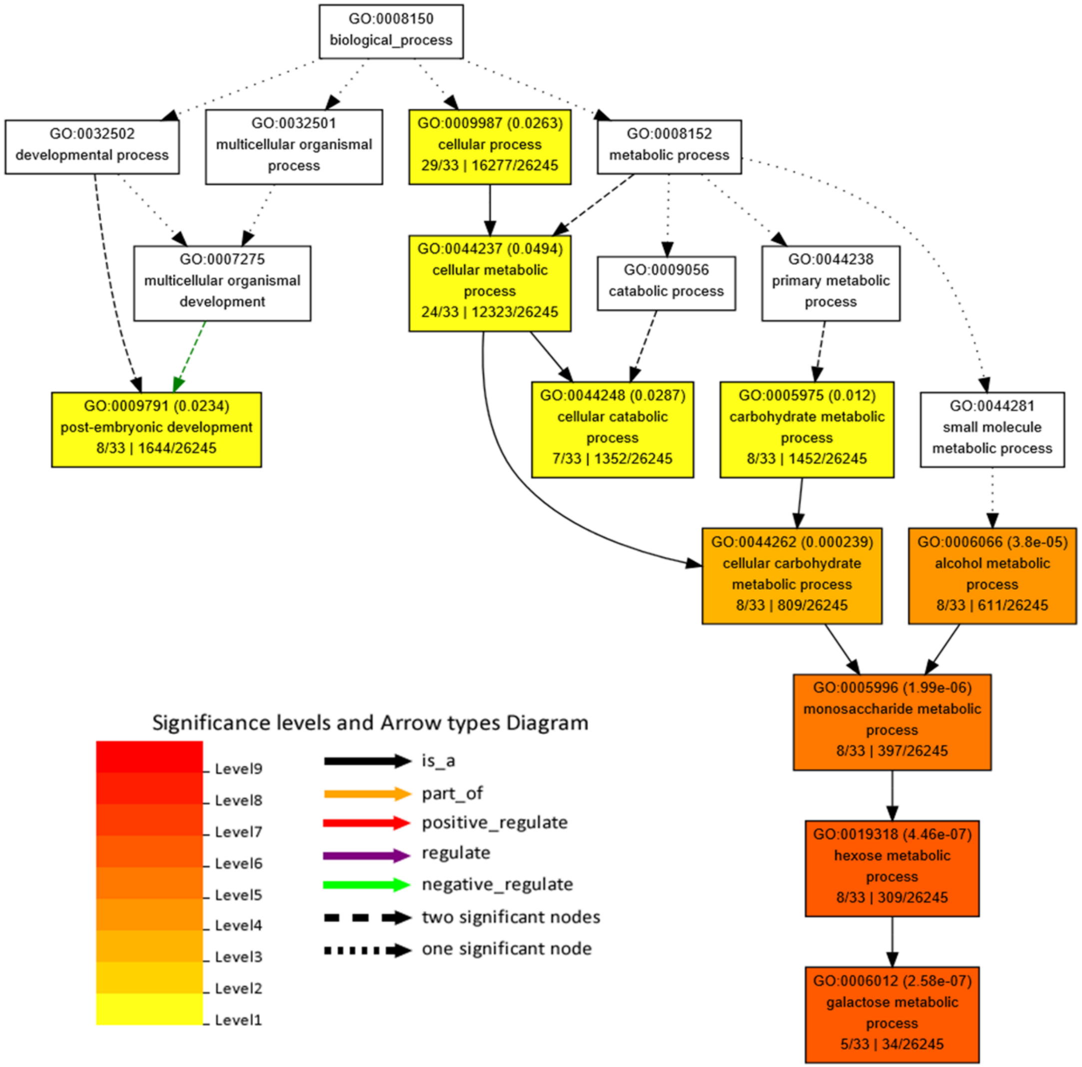
**缺点:**

1. 许多富集的GO term含义比较笼统，如transcription, cell cycle，不能帮助更好地聚集研究线索。
   – 这是因为针对2x2列联表的统计检验，如Fisher‘s test、Chi-square test，如果把列联表的四个数都乘以10，富集比例不变，但p-value会变得更显著（**课后练习**）。这些笼统的GO term集合包含的基因数目多，它们就容易得到显著的p-value。

2. GO terms组成一个类似树状的关系图，terms沿树根到枝叶是从笼统到具体的功能关系，在树中临近的GO terms含义相关，也包含许多共同的基因，所以富集结果中会看到冗余的GO terms富集。

3. 图比较普通，缺少新奇感。

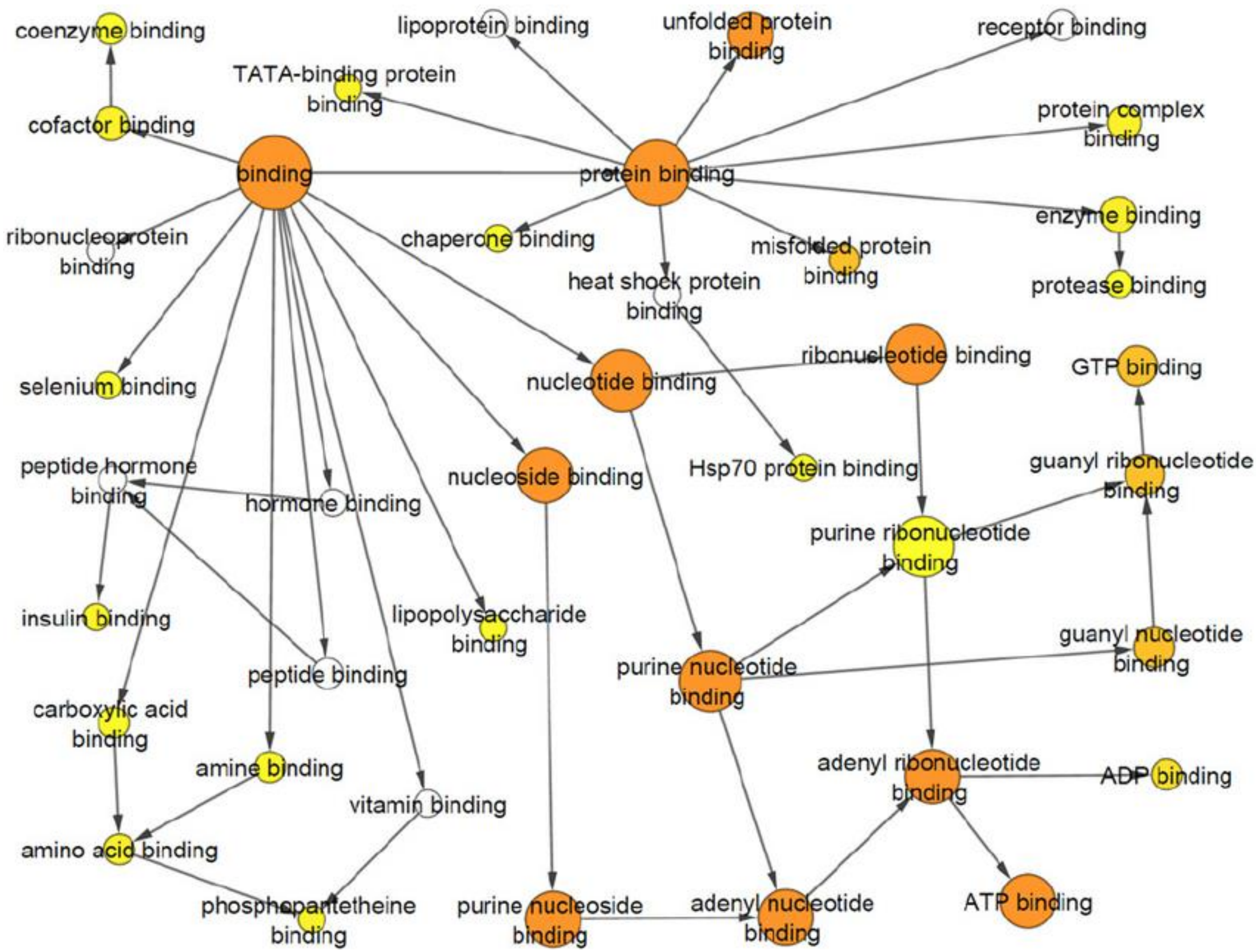
# GO analysis using AgriGO




改进1：把GO term的关系树画出来，富集的程度用颜色表示，很直观地看到相邻term富集程度的渐变

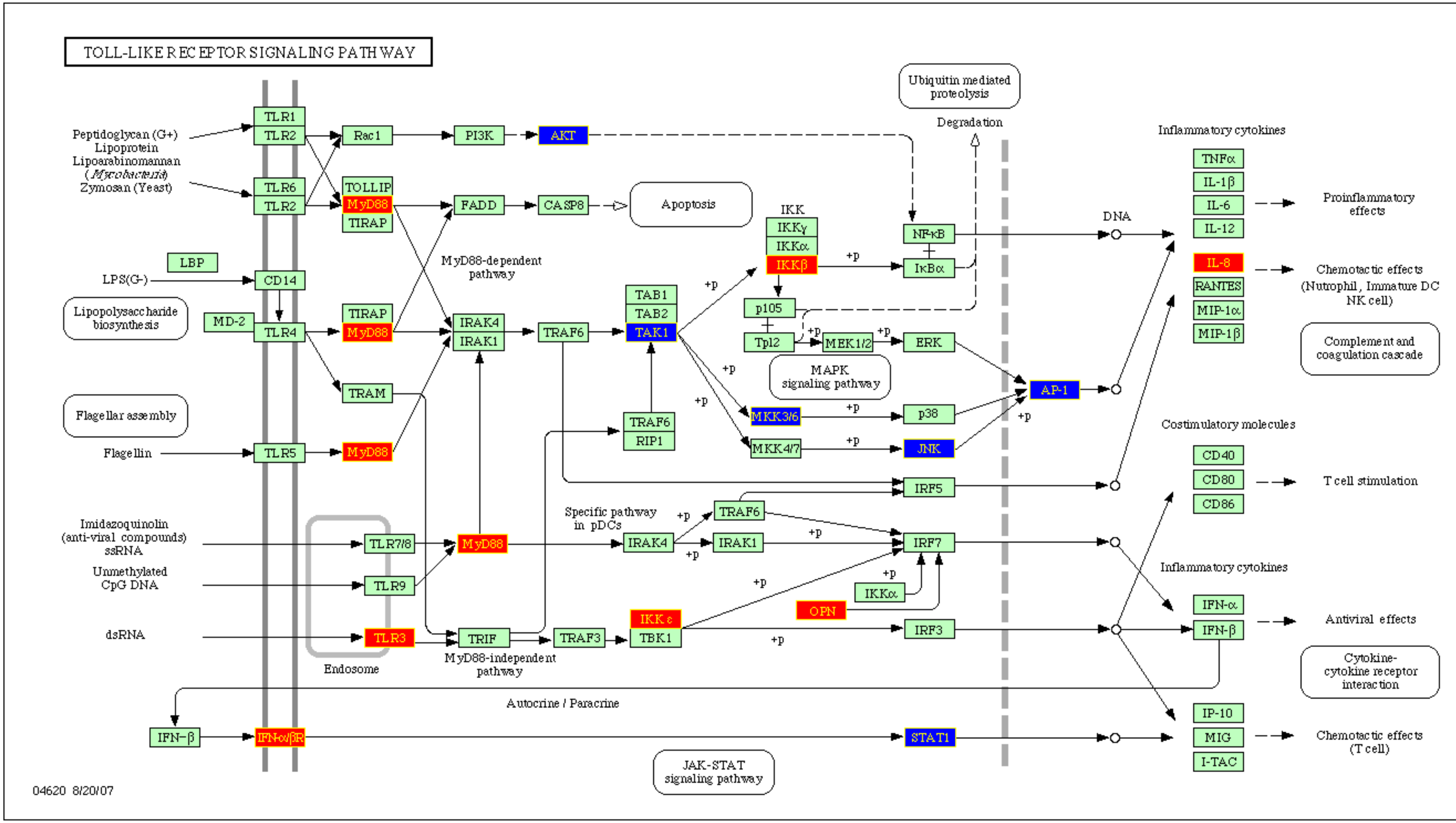
https://figshare.com/articles/_Gene_Ontology_GO_analysis_of_the_candidate_genes_using_AgriGO_/1418715

# Network plot




在平面图中用颜色、大小表示更多维度的信息：用圆的大小表示GO term包含的基因个数

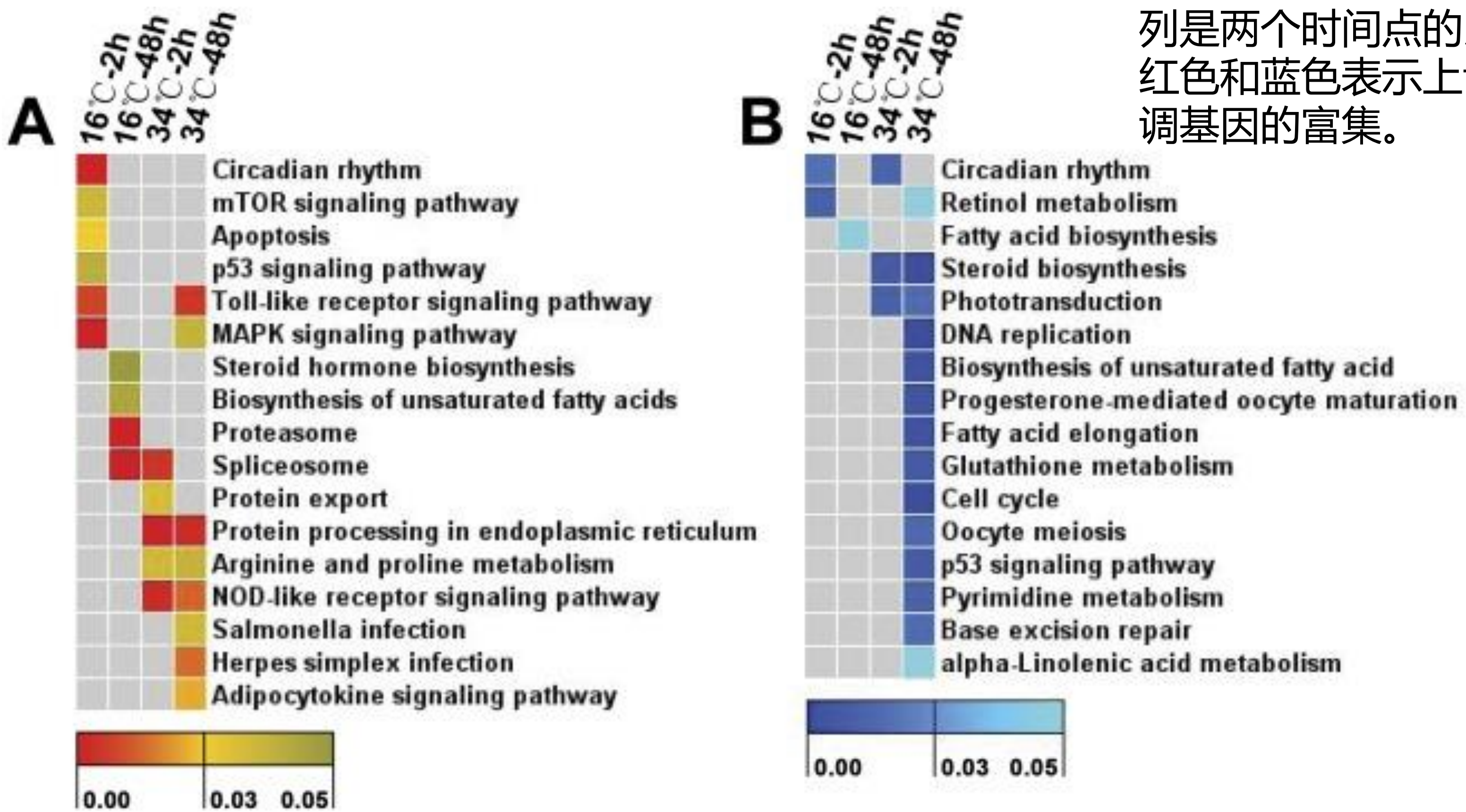
# KEGG plot




改进2：在感兴趣的或富集分析得到的生物通路图（如KEGG 通路，https://www.genome.jp/kegg/ ）中，用颜色标出基因表达变化，帮助看到变化的通路节点或分支

# 多条件下的富集热图

这里每行是不同通路，每列是两个时间点的比较，红色和蓝色表示上调和下调基因的富集。

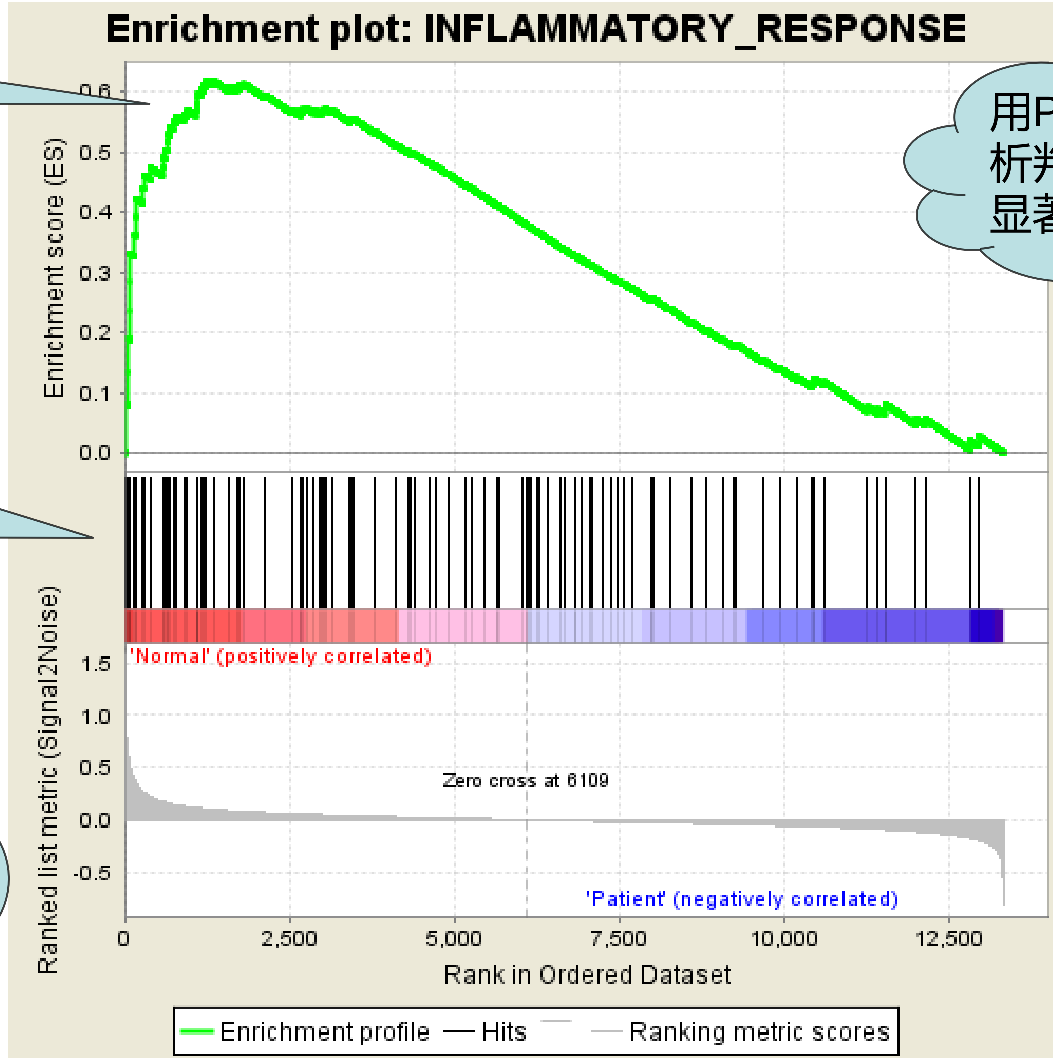



改进3：相比GO terms，生物通路数量少一些，相互也更独立些，所以可以用热图同时展示多个通路在不同条件下差异表达基因的富集情况，看到通路富集随时间的变化

# Gene set enrichment analysis (GSEA)

富集打分

用Permutation分析判断富集打分的显著性




感兴趣的基因功能/通路列表，如细胞周期基因

**所有**基因的**排序**，例如按差异表达

**优点**：不用特定差异表达阈值，使用所有基因差异表达的信息

PNAS 2005 Gene set enrichment analysis A knowledge-based approach for interpreting genome-wide expression profiles

软件主页：http://www.broadinstitute.org/gsea/

算法介绍（中文）：http://blog.csdn.net/qq_29300341/article/details/52956052

# 文献举例

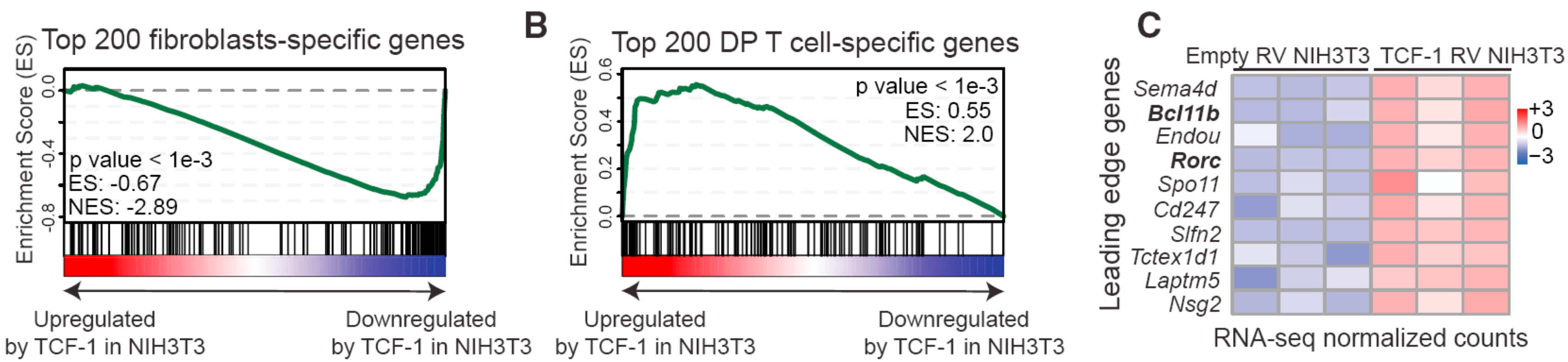


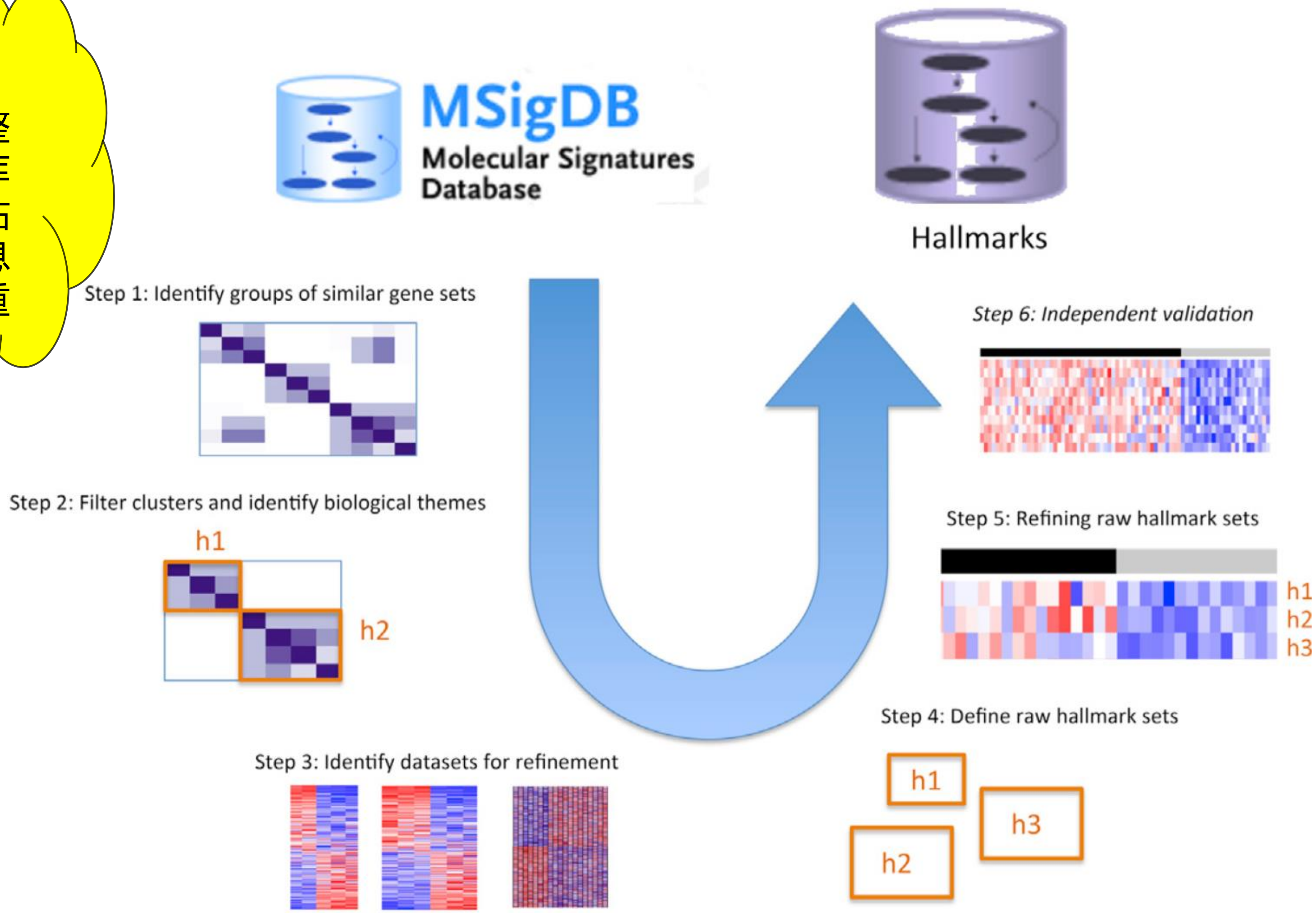

**Figure 6. T Cell-Specific Genes Innately Repressed in Fibroblasts Are Upregulated by TCF-1 (see also Figure S6)**

(A–C) Three replicates of RNA-seq in TCF-1 RV and Empty RV NIH 3T3 cells were generated to assess the effects on gene expression. DESeq2 (fold-change > 2 and p adj < 0.05) facilitated the differential gene-expression analysis resulting in 1,295 down- and 1,477 upregulated genes in TCF-1 RV NIH 3T3 cells (see Figures S6A and S6B). Genes located in non-canonical chromosomes were removed from the lists. In addition, we applied differential gene-expression analysis between Empty RV and DP T cells to establish cell specific gene expression (see STAR Methods), which facilitated GSEA analysis of DEGs in TCF-1 RV NIH 3T3 cells on the fibroblast gene set (A) and the T cell gene set (B). Leading edge analysis (C) in top T cell genes.

Immunity 18 Lineage-Determining Transcription Factor TCF-1 Initiates the Epigenetic Identity of T Cells

# The Molecular Signatures Database Hallmark Gene Sets

**生信思路**：
数据收集整理、数据库和分析网站是生物信息学的一个重要分支




Generated 50 “hallmark” gene sets from the Molecular Signature Database (MSigDB).
The hallmarks reduce redundancy and produce more robust enrichment analysis results.

*Cell Systems* 2015 The Molecular Signatures Database Hallmark Gene Set Collection

# 本节内容

- 分类数据举例
- 基因功能富集分析
- 分类数据的统计**检验**
- 统计**陷阱**：辛普森悖论
- 统计**作图**：富集作图
- 富集分析软件

# 用网站做分析：基因富集分析

c-Myc

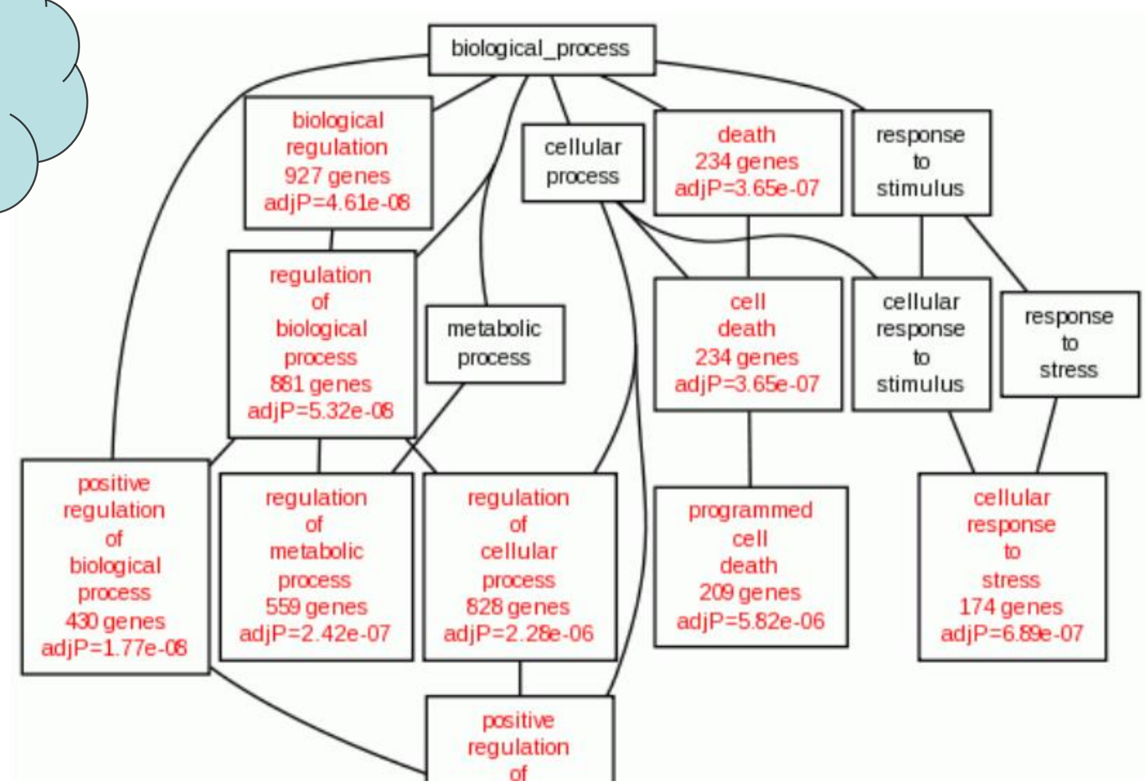
WebGestalt

WEB-based GEne SeT AnaLysis Toolkit

*Translating gene lists into biological insights...*

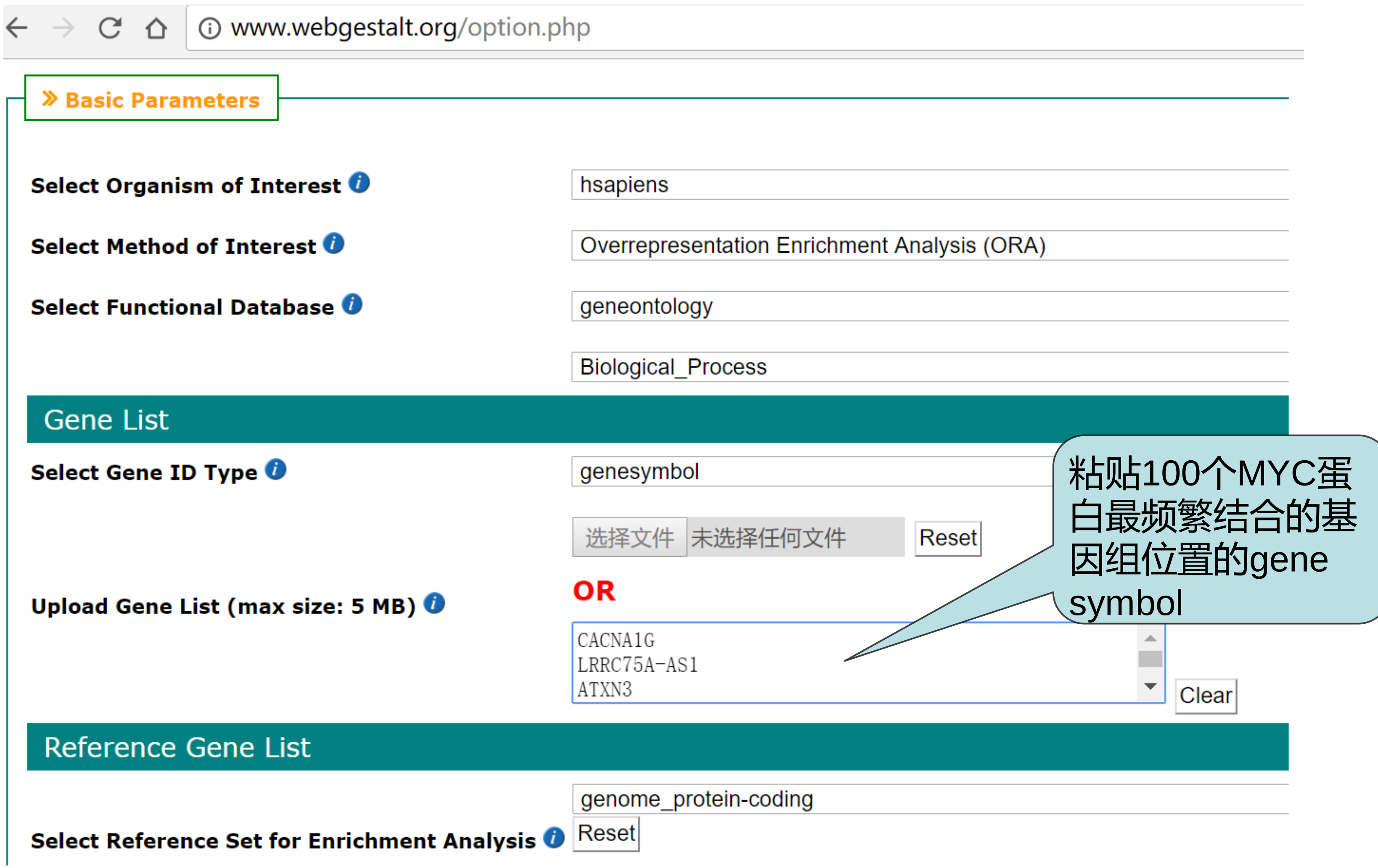
| |
|---|
| CACNA1G |
| LRRC75A-AS1 |
| ATXN3 |
| ST3GAL3 |
| DISC1 |
| PGAP2 |
| PLAGL1 |
| MUC1 |
| PDE9A |
| UBE2V1 |
| VEGFA |
| BCL2L11 |
| ZMYND8 |
| CTCFL |
| CHN2 |
| IKZF3 |
| ZNF133 |
| ZNF302 |

**问题**：MYC蛋白可能调控哪些基因通路?




MYC 结合位点基因
(MYC binding.xlsx)

# WebGestalt 演示

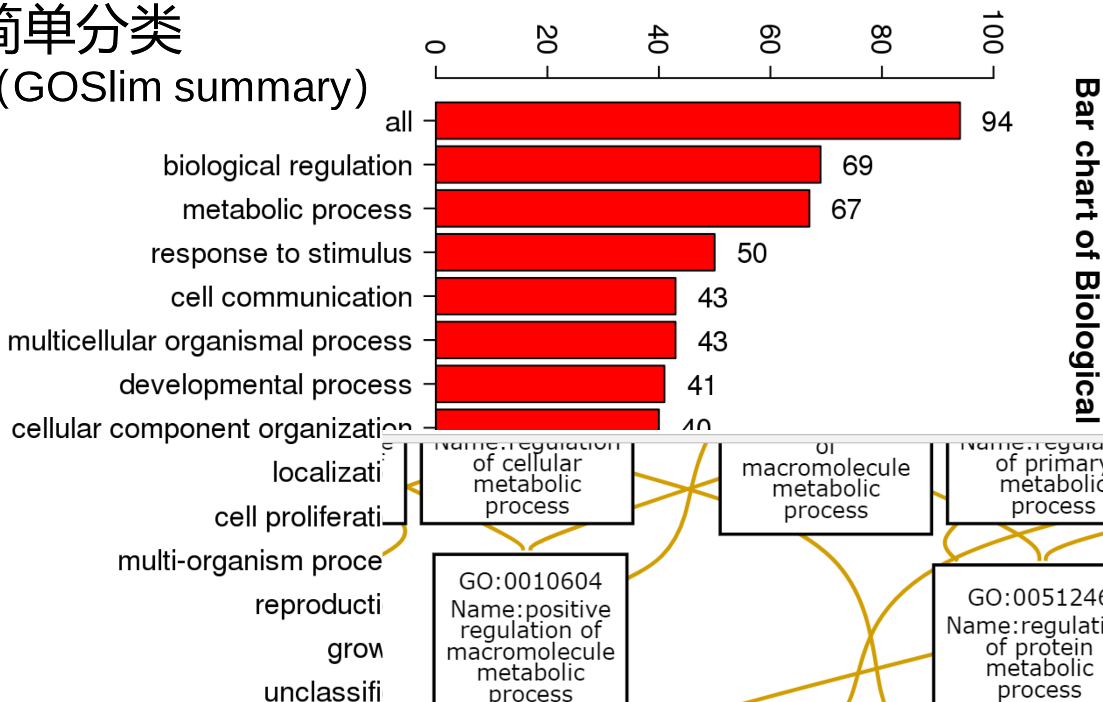


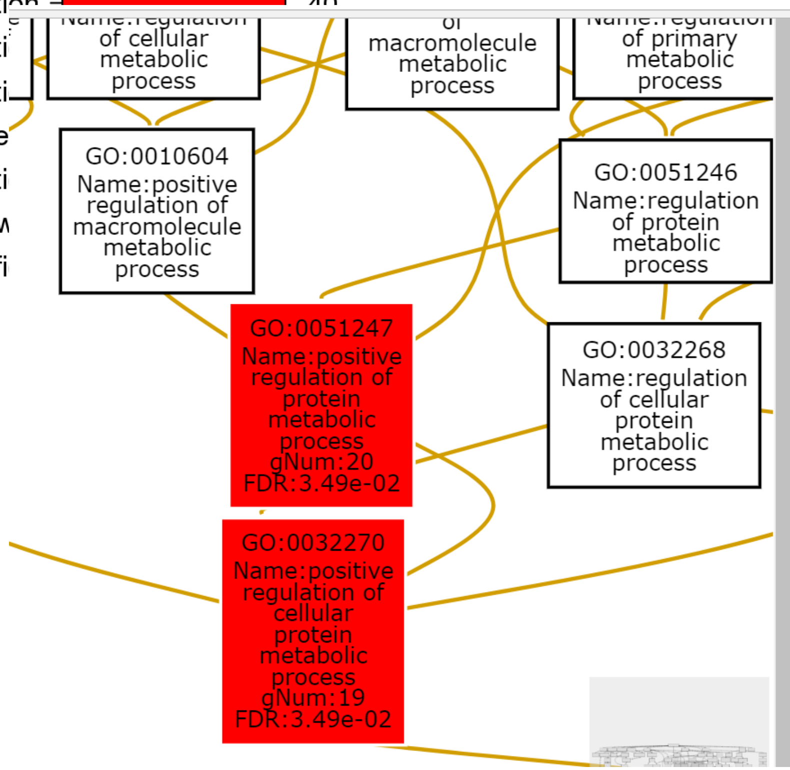

http://www.webgestalt.org/

另一个常用基因功能富集网站：DAVID，https://david.ncifcrf.gov/

# WebGestalt 分析结果

简单分类
（GOSlim summary）




**82** input IDs and **16258** reference IDs are annotated

Enrichment Results

- C: GO category size
- O: observed genes
- E: Expected genes
- R: Enrichment ratio

**ID:GO:0051247 Name:positive regulation of protein metabolic process**

**C=1485; O=20; E=7.03; R=2.84; PValue=1.32e-05; FDR=3.49e-02**

| userid | Gene Symbol | Gene Name | Entrez Gene |
|---|---|---|---|
| ABI1 | ABI1 | abl interactor 1 | 10006 |
| BCL2L11 | BCL2L11 | BCL2 like 11 | 10018 |
| ANAPC10 | ANAPC10 | anaphase promoting complex subunit 10 | 10393 |
| GLIPR2 | GLIPR2 | GLI pathogenesis related 2 | 152007 |
| DAG1 | DAG1 | dystroglycan 1 | 1605 |

# Pathway/KEGG富集分析

| ID | Name | #Gene | FDR |
|---|---|---|---|
| hsa01521 | EGFR tyrosine kinase inhibitor resistance - Homo sapiens (human) | 3 | 7.32e-01 |
| hsa05166 | HTLV-I infection - Homo sapiens (human) | 5 | 7.32e-01 |
| hsa04215 | Apoptosis - multiple species - Homo sapiens (human) | 2 | 7.32e-01 |
| hsa04914 | Progesterone-mediated oocyte maturation - Homo sapiens (human) | 3 | 7.32e-01 |
| hsa04933 | AGE-RAGE signaling pathway in diabetic complications - Homo sapiens (human) | 3 | 7.32e-01 |
| hsa05340 | Primary immunodeficiency - Homo sapiens (human) | 2 | 7.32e-01 |

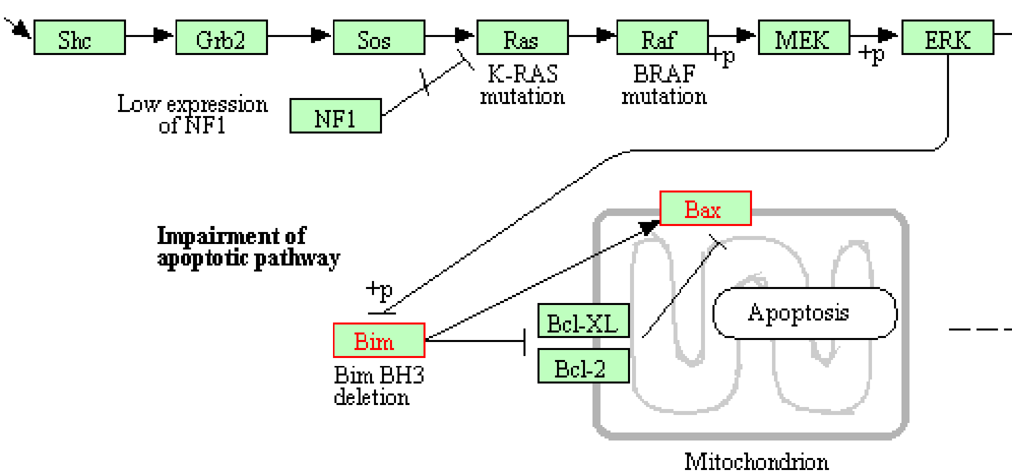
ID Search

Download Table

**ID:hsa01521 Name:EGFR tyrosine kinase inhibitor resistance - Homo sapiens (human)**

**C=81; O=3; E=0.43; R=6.99; PValue=8.81e-03; FDR=7.32e-01**

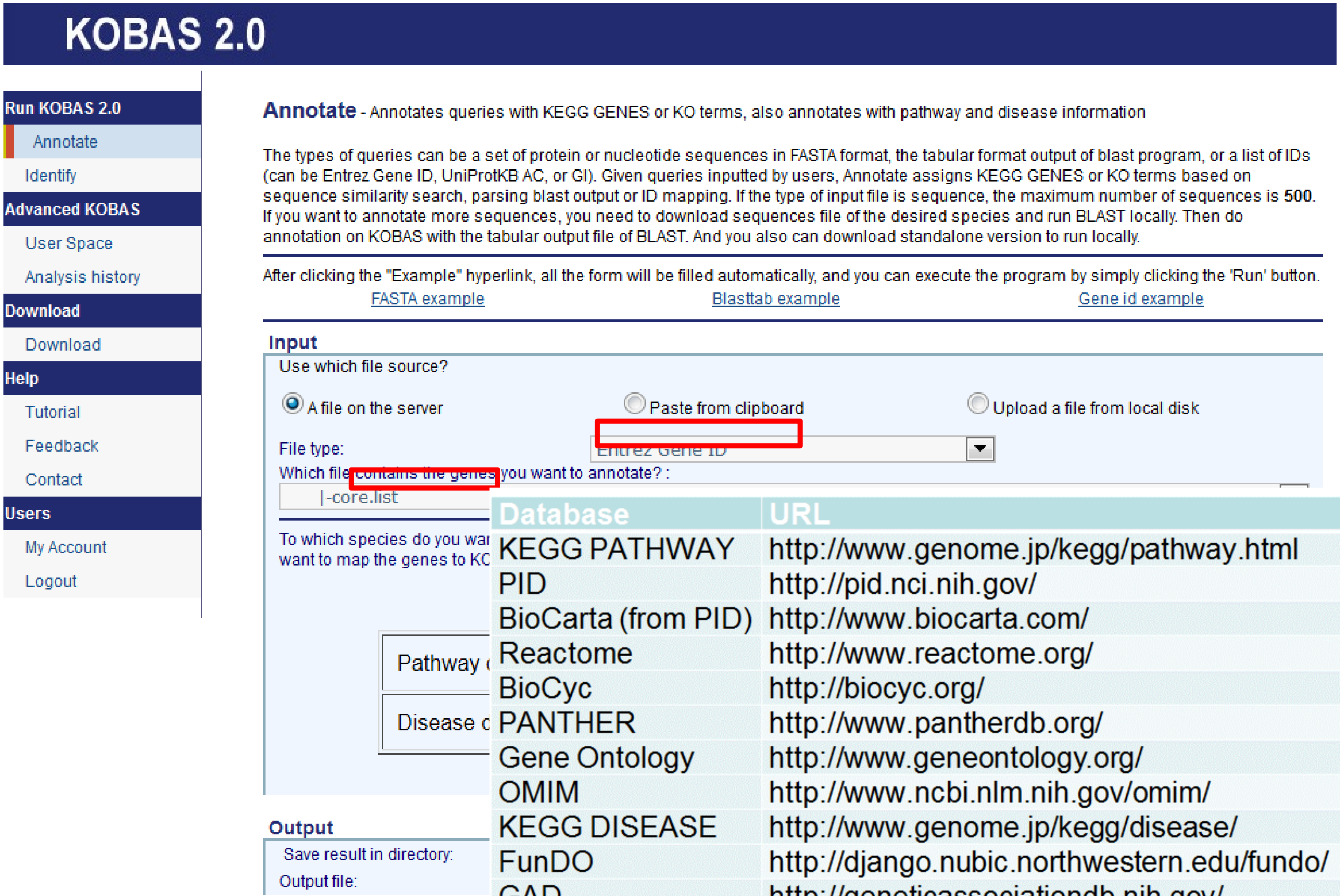
| userid | Gene Symbol | Gene Name | Entrez Gene |
|---|---|---|---|
| BCL2L11 | BCL2L11 | BCL2 like 11 | 10018 |
| BAX | BAX | BCL2 associated X, apoptosis regulator | 581 |
| VEGFA | VEGFA | vascular endothelial growth factor A | 7422 |




**生信思路：**

- 关注预期和意料之外的富集通路
- 关注输入基因列表和通路的交集基因（通路图中标为红色）、它们在通路中的上下游关系（富集分析中没有用到网络结构信息）

# KOBAS web server




| Database | URL |
|---|---|
| KEGG PATHWAY | http://www.genome.jp/kegg/pathway.html |
| PID | http://pid.nci.nih.gov/ |
| BioCarta (from PID) | http://www.biocarta.com/ |
| Reactome | http://www.reactome.org/ |
| BioCyc | http://biocyc.org/ |
| PANTHER | http://www.pantherdb.org/ |
| Gene Ontology | http://www.geneontology.org/ |
| OMIM | http://www.ncbi.nlm.nih.gov/omim/ |
| KEGG DISEASE | http://www.genome.jp/kegg/disease/ |
| FunDO | http://django.nubic.northwestern.edu/fundo/ |
| GAD | http://geneticassociationdb.nih.gov/ |
| NHGRI GWAS Catalog | http://www.genome.gov/gwastudies/ |

北京大学生科院魏丽萍组，http://kobas.cbi.pku.edu.cn/

# 法则 4：思考备选解释
# 举例：富集分析的背景基因选择

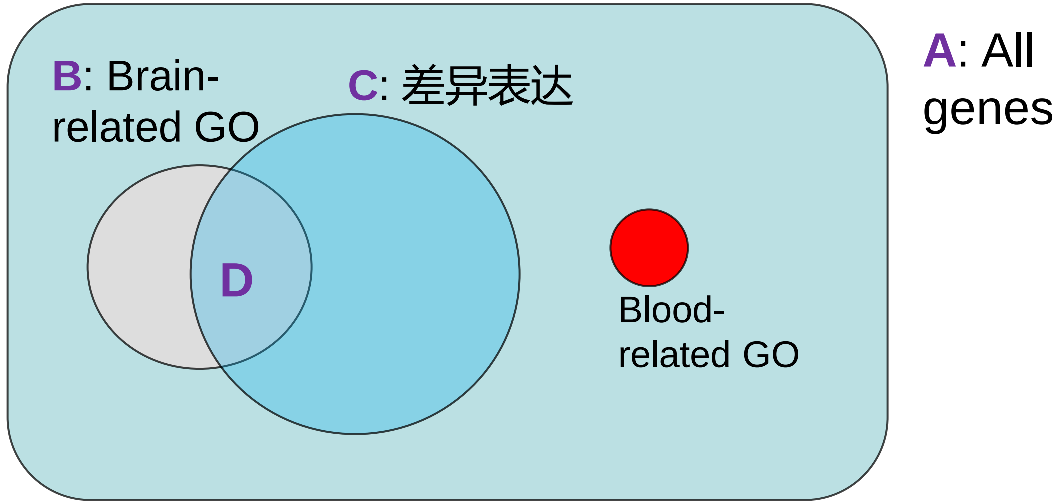
分析功能基因组的数据时，我们经常会得到一个人类基因列表，比如在正常和癌症细胞里有差异表达的基因。一个一个看这些基因的已知功能的描述太繁琐，也不容易看出规律。一个常用的方法是Gene Ontology的**富集**分析（比如https://david.ncifcrf.gov/）。许多基因的功能可以用一些规范的条目（GO terms）表来描述，这个分析方法就能找出我的基因列表中富集的GO条目。

**下面两种分析方法你认为那个更合理呢：**

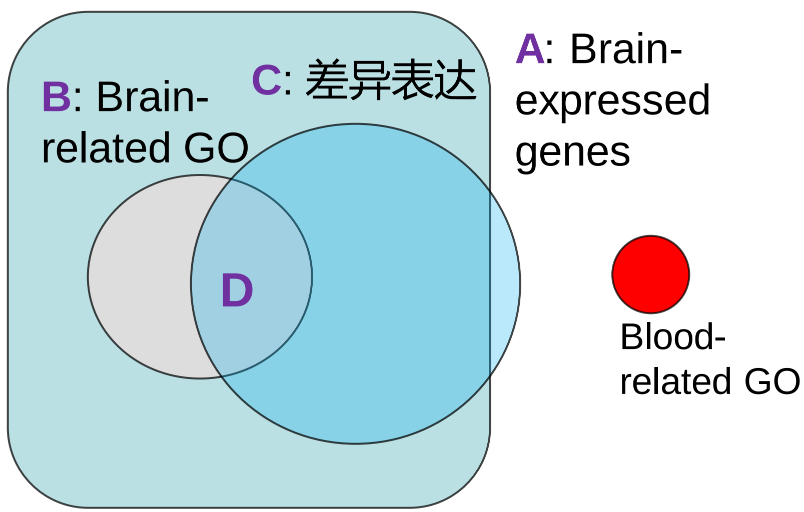
1）直接提交我的基因列表，这个网站会用所有人类基因作为比较的背景，使用超几何分布来找出显著富集的GO条目。这些功能就可能是和我在研究的癌症相关的。许多论文不是这么做的吗？

2）假如我的差异表达数据是关于大脑的疾病，因为大部分有差异表达的基因也在大脑细胞中表达，即使我的基因列表和这种疾病没有关系（比如随机从在脑细胞中表达的基因中选取一个子集），它也会富集一些GO条目（比如和脑功能相关的）。所以，要在富集分析的网站上同时提交一个脑细胞中表达的基因列表作为“Reference gene list”来校正这种影响。

# 富集分析中背景基因的影响




**分析思路**（刘玉婷）：分别看上调和下调的基因中富集的通路，是否有异同?




- **A:** 列联表分析考虑的所有基因
- **C:** 上页中差异表达的基因，大都是大脑表达基因
- 富集分析比较的比例：**D/C vs. B/A**

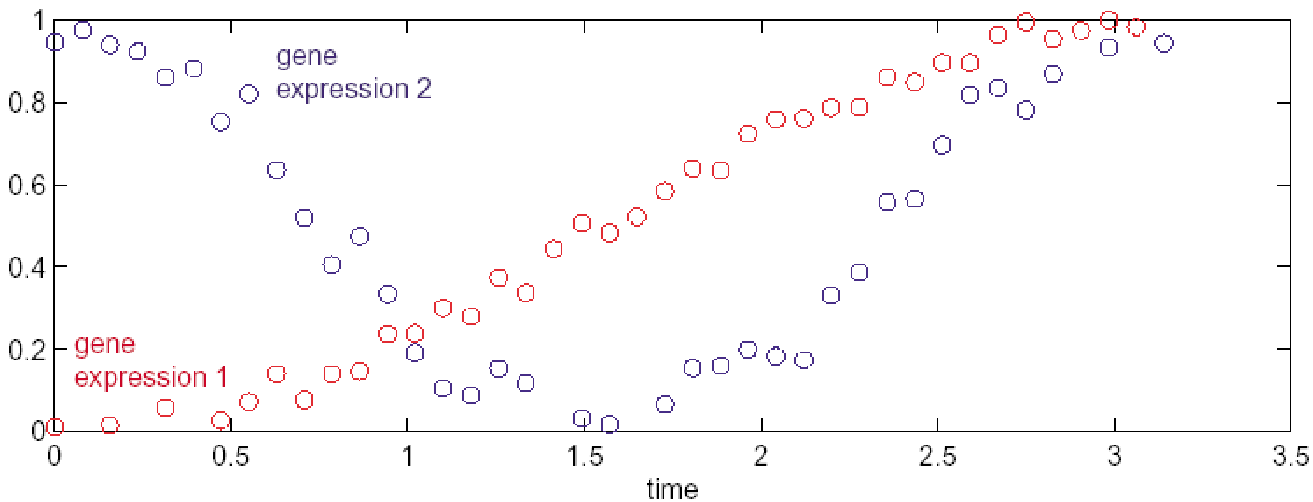
# 课后练习

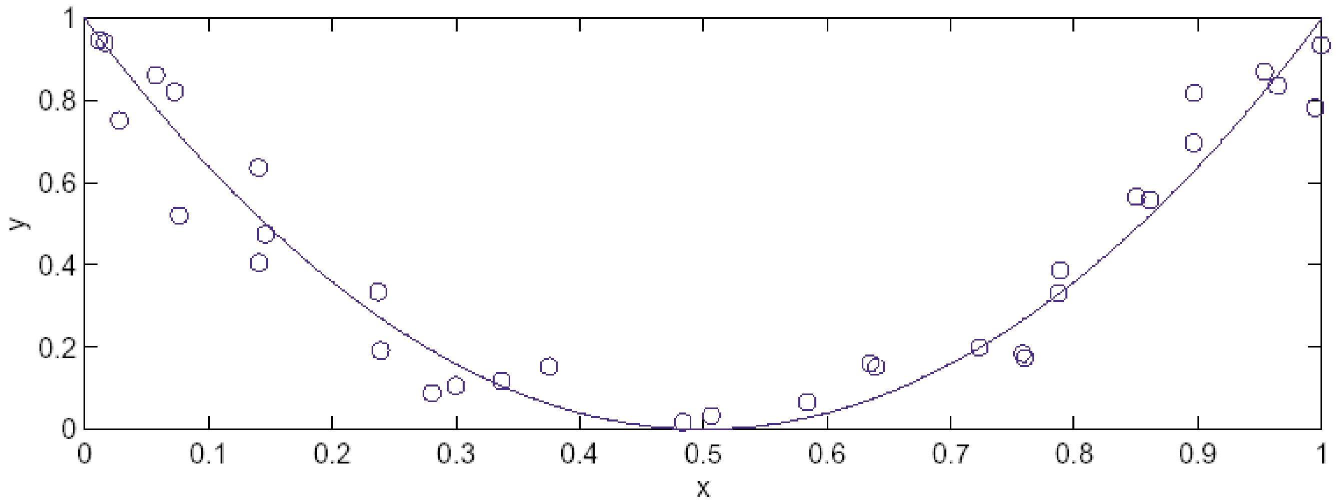
1. 阅读文献《Immunity 18 Lineage-Determining Transcription Factor TCF-1 Initiates the Epigenetic Identity of T Cells》，哪些图用到了基因富集分析？显著性富集提示了什么样的生物功能?

2. 查看R语言相关函数（“R functions for count data”页）的参数和例子（例如用?fisher.test）。使用相关函数检验“比例的比较：TCF-1”例子(p.13)中的频率差异是否显著?

3. 上网搜索(如www.bing.com)，发现使用网站或R语言做基因功能富集、GSEA分析的工具和教程。练习使用WebGestalt或其他工具，对MYC 靶基因列表（MYC binding.xlsx）做富集分析。

# 课后练习4：How to detect nonlinear relationship?




提示：将X、Y数值转换为高、中、低三类，做3x3列联表